

Making Information Stick: Goal Setting and Attention

Abstract

Organizations are increasingly held accountable for the emissions their customers generate in using their products, yet the instrument they most often deploy — *information provision* — tends to raise awareness without changing behavior. We argue that this behavioral issue reflects limited attention: individuals attribute low instrumental value to the information and stop attending to it. Pursuing a goal, by contrast, may increase the perceived value of that information, sustaining attention and improving recall. We test this idea in a pre-registered field experiment (N=830), with a major European telecommunications operator, on *offloading* – i.e., switching internet use from mobile to Wi-Fi networks – which lowers the emissions of data use. While information provision alone does not produce any significant change in consumption behavior, the addition of goal setting increases attention in the short run and recall in the long run. These effects together increased the Wi-Fi share of total data consumption by about five percentage points (0.18 s.d.), without altering total data use. This behavioral shift towards cleaner networks persists after the intervention ends. Evidence from a follow-up survey indicates that this change is not due to alternative mechanisms, such as automated data management technologies, thereby lending support to theories of rational inattention. Self-set goals offer firms a low-cost, scalable design feature to reduce use-phase emissions without curtailing the service customers value.

Keywords: goal setting, inattention, information, memory, digital technologies, pro-environmental behavior

JEL Codes: C93, D91, D83, Q50

1. Introduction

Organizations are increasingly held responsible not only for their own environmental footprint but for the emissions their customers generate in using their products (Hertwich & Wood, 2018). In response, they routinely deploy interventions that aim to shape how consumers use their products and services. A primary barrier to optimal consumption choices is imperfect information about product characteristics and benefits (Gao & Tavoni, 2023). Knowledge gaps can reduce the demand for high-quality products, resulting in suboptimal consumption decisions that yield private and social losses. Informational barriers are most damaging when they lead consumers to overuse carbon-intensive options, whose emissions are borne by society at large and exacerbate climate change. Because greenhouse gas emissions are long-lived, organizations should think of interventions that shift consumption behavior lastingly, not merely while in place.

To correct misperceptions and ignorance, firms and policymakers have long relied on informational instruments, such as labels or awareness campaigns (Cason & Gangadharan, 2002; Allcott & Knittel, 2019; Allcott & Taubinsky, 2015). Yet, the effectiveness of such interventions is often limited (Rodemeier & Löschel, 2025) and cannot be explained solely by a lack of knowledge. Individuals face attention constraints, and like any scarce resource, its optimal allocation may involve rational inattention to some piece of information (Gabaix, 2014, 2019; Gabaix & Laibson, 2006; Golman et al., 2017; Sims, 2010). Moreover, even when initially attentive, individuals tend to forget or devalue information over time as salience fades or cognitive load increases (Graeber et al., 2024; Kahana, 2012). Consequently, information provision frequently raises awareness without producing behavioral change (Benartzi & Thaler, 2007; Dechezleprêtre et al., 2025; Milkman et al., 2021; Vlasceanu et al., 2024). In this work, we show that goal setting increases the perceived instrumental value of information, which in turn improves attention and recall, leading to lasting behavioral changes.

Goal setting has been proposed as a way to overcome inattention by increasing the perceived *instrumental value* of information (Bordalo et al., 2022; Dijksterhuis & Aarts, 2010; Robison et al., 2021). When individuals define and work toward a goal, the information relevant to achieving that goal becomes more salient and more likely to be encoded in memory.¹ Goal setting transforms passive information into a *strategic input* for personal decision-making: individuals pay more attention to, recall better, and act upon information that serves the pursuit of an objective. In economic terms, a goal provides a reference point for self-evaluation, creating psychological costs for deviations (Bryan et al., 2010; Hsiaw, 2013; Locke & Latham, 2019). This increases the marginal utility of attention, possibly mitigating inattention and memory decay (Gabaix & Laibson, 2006; Sallee, 2014; Schwartzstein, 2014). Yet the core claim, that goals work by directing attention and protecting memory, has rarely been isolated and measured directly, especially in the field.

We provide field experimental evidence on how goal setting enhances the effectiveness of informational policy instruments by directing attention and improving memory. Our study focuses on a growing yet understudied source of environmental externalities: the energy consumption associated with mobile data transmission. Specifically, we examine *offloading* behavior – the action of switching internet use from mobile to Wi-Fi networks – which can reduce the per-byte CO₂ emissions by up to 75% (ADEME, 2022). Because offloading behavior is a new context characterized by limited awareness among

¹ There is a growing literature at the cross of economics and psychology that recognizes attention as a limited resource and hence selective: the mind selects what we focus on and what to ignore. Two broad selection mechanisms have been put forward top-down and bottom up. While bottom-up mechanisms are stimulus-driven processes in which attention is captured by features that are externally salient (e.g., contrast, novelty, intensity), top-down mechanisms are goal-directed processes in which attention is guided by the decision-maker's goals, objectives, and motivations (Bordalo et al., 2022; Dijksterhuis & Aarts, 2010). In economics, models of rational inattention (Sallee, 2014; Sims, 2010) and sparsity (Gabaix, 2014) formalize this selective allocation of attention as an optimization problem: decision-makers have rational expectations and choose what information to process given cognitive constraints and perceived payoffs. By prompting individuals to define explicit goals, goal pursuit can shift these attention weights, raising the perceived instrumental value of certain information and thus mitigating inattention and belief distortions at their source.

consumers (Figure A1), it provides an ideal setting to test information provision, and whether goal setting increases its salience and persistence. We partnered with a major European telecommunications company to design a randomized controlled trial with over 800 smartphone users. Participants tracked their mobile and Wi-Fi data consumption for several weeks and were randomly assigned to one of three conditions: (i) a baseline group with no intervention, (ii) an *Information* group receiving personalized feedback on the CO₂ emissions of their data use, and (iii) a *Goal Setting* group that also had the option to set weekly self-imposed goals for reducing mobile data use.

Our findings reveal three main results. First, information provision alone does not significantly alter data consumption behavior: participants informed about the relative emissions of mobile and Wi-Fi networks did not increase their offloading to Wi-Fi. Second, adding goals significantly increased the share of Wi-Fi in total data use—by about five percentage points relative to the control—without changing total data consumption. This indicates a substitution of cleaner over dirtier technologies rather than a reduction in total usage. Third, these effects persisted for at least two weeks after the end of the intervention and were sustained by higher *information recall* even six months after the end of the study. Participants who set goals not only shifted their behavior but also retained more accurate information about the environmental cost of mobile data, suggesting that goal setting increases informational recall.

We provide two sources of evidence of the mechanisms explaining the difference between *Information* and *Goal Setting* conditions. We first explore the role of attention using the time that participants spent interacting with the information provided. While participants in both conditions spent the same amount of time on the informational page the first time they saw it, participants in the *Information* condition spent significantly less time in the following weeks. Participants in the *Goal setting* condition did not show a similar decline. Second, data from a pre-registered follow-up study suggest that goal setting leads to more accurate information recall. Put differently, goal setting led to sustained behavior change by increasing attention in the short term and information retention in the long term. Finally, we also rule out a different technology-based mechanism (Brandon et al., 2017) by investigating whether participants assigned to our experimental interventions changed some smartphone settings to facilitate automatic offloading. We find no significant differences in any of these measures across conditions that suggests this mechanism.

Our work contributes to the current debate on the role of attention and memory in decisions (Bordalo et al., 2020; Graeber et al., 2024). Economics models of inattention (Gabaix, 2019; Schwartzstein, 2014) view it as a scarce resource allocated toward information with the highest perceived payoff. In these frameworks, individuals do not fully process all available information but instead prioritize signals that are instrumentally valuable for decision-making. We propose that goal setting raises the perceived instrumental value of specific information, effectively shifting attention weights toward goal-relevant signals. This mechanism aligns with evidence from psychology and neuroscience (Desimone & Duncan, 1995; Dijksterhuis & Aarts, 2010) that explicit goals guide top-down attention, and with economic models in which the value of information depends on expected utility gains (Sallee, 2014; Sims, 2010). By making relevant information more valuable to process, goal setting reduces inattention at the moment of exposure and improves retention, helping information interventions overcome one of their central limitations.

We also relate to the literature studying the effect of informational strategies. Evidence on the effectiveness of information programs is mixed across domains (Allcott & Knittel, 2019; Allcott & Taubinsky, 2015). These programs typically rest on the premise that behavior improves once knowledge gaps are filled and individuals are made aware of the relevant course of action (Gao & Tavoni, 2023). However, individuals continue to make sub-optimal decisions even when valuable and actionable information is provided (Handel & Schwartzstein, 2018). Our contribution is to show that limited attention is a key factor leading to these weak effects. We provide evidence that introducing self-set

goals increases the saliency of information in the short term (i.e., individuals pay more attention to it), and improves recall in the long term, effectively addressing the knowledge gaps that information alone often fails to close. These results are complementary to recent field work showing that paying individuals for attention improves decision-making and welfare (La & Myers, 2026).

Our study also relates to the vast literature on goal setting (Locke & Latham, 2002, 2019) by contributing in two ways. First, goal-setting strategies have not received much attention in the field and, to the best of our knowledge, the existing evidence comes from studies that have bundled the two interventions together without isolating their marginal effect (Abrahamse et al., 2007; Loock et al., 2013; Löschel et al., 2023). Our experimental design allows to discern the relative advantages of goal setting versus the simple provision of information, and to look at the interplay between the two.

Second, it proposes attention and memory as mechanisms behind goal setting. This attentional channel is not an alternative to the reference-dependent view of goal setting (Harding & Hsiaw, 2014), in which a goal acts as a reference point that motivates effort directly. These two are complementary: a goal can raise the marginal value of attending to relevant information precisely because it creates a reference point against which behavior is evaluated. Our contribution is to make the attentional and mnemonic step observable, documenting that goal setters sustain engagement with information and retain it longer, rather than claiming that reference-dependent motivations are the ultimate driver of behavior.

Our study also adds to the recent debate on the persistence of behavioral solutions in the context of resource conservation (Bernedo et al., 2014; Brandon et al., 2017). Whether the effect of interventions intended to change behavior persists after the intervention stops is an open question in the literature (Brandon et al., 2017; Frey & Rogers, 2014; Vivek et al., 2021). Persistence arises from fostering good habits, shaping beliefs, and harnessing external reinforcement. Our results are in line with recent field works finding a long-lasting effect of goals (Alonso-Pauli et al., 2022), and complementary to all studies showing short-lived effects of pure information-based interventions (Andor et al., 2022).

The remainder of the paper is organized as follows. Section 2 describes the experimental design and data collection. Section 3 reports the results of the experiment. Section 4 discusses the possible mechanisms. Section 5 concludes.

2. Study Design

Context. We partnered with a large European telecommunications company (*Orange*) to co-develop interventions in *mobile data offloading*, i.e., switching smartphone data use from 4/5G to Wi-Fi. Data centers and communication infrastructures generate a share of global emissions comparable to the whole aviation industry (Royal Society, 2020). Recent estimates show that mobile networks (4/5G) can produce up to 4 times the CO₂ quantities of Wi-Fi networks per gigabyte (Gb) consumed (ADEME, 2022). This setting poses major challenges for firms, which face growing pressure to measure and reduce the emissions associated with customers' data use, and for European consumers, where unlimited mobile plans make the marginal cost of mobile data close to zero. Yet public awareness of the issue remains limited. In a large customer survey, 53% report believing that digital technologies have a positive environmental impact (Orange, 2021). Consistently, our pre-experimental survey (Figure A1) reveals two stylized facts. First, 43% of respondents incorrectly believe Wi-Fi is as polluting as (or more polluting than) mobile data. Second, about 40% state a preference for mobile data even when free Wi-Fi is readily available. Those preferring mobile are more impatient and present-biased, suggesting that offloading decisions may be shaped by temptation in addition to misperceptions.² Put together, the

² We measured participants time preferences using a self-declarative question from Falk et al., 2023 (see Appendix D). Moreover, our elicited measure of time preferences significantly correlates with pre-experimental levels of offloading (Pearson $r = 0.166$, $p = 0.006$) suggesting that more patient participants predominantly use more Wi-Fi networks than mobile ones in their data consumption.

presence of wrong priors and consumers' impatience makes the offloading context a perfect testbed for studying information provision and goal setting.

Design. We designed a framed field experiment (Brent et al., 2017; Harrison & List, 2024) with university students from 4 different large campuses in France (Dijon, Montpellier, Nice, and Strasbourg). Student populations represent the user category with the highest level of data consumption, due to high social media use and video streaming (Open Signal Report; Fogg, 2018). This study was pre-registered (Aspredicted #110437) and all materials are available in a public repository (https://researchbox.org/1325&PEER_REVIEW_passcode=TZXHQN). Minor deviations and additional analyses are reported and discussed in the Appendix (see Appendix G) and do not affect the main experimental results. Minimum required sample size was pre-determined following a power analysis (see Appendix F).

Data collection strategy. We collected data on mobile and Wi-Fi data consumption over an 8-week period (Figure 1). To take part in the study, participants had to clear a screening survey (*Recruitment Survey*) which asked demographic questions and screened out all non-Android users. Participants passing this initial survey were asked to download a free app (*GlassWire*; see Appendix C) available on all Android devices and reply to an online questionnaire sent at the beginning of each week of the study (henceforth, *Reports*). Because the app is owned by third parties, we could not use it to directly gather consumption data. Yet, it provided participants with a means to access their consumption levels which they had to report each week onto our experimental web platform accessible via a personalized link and coded in oTree (Chen et al., 2016; see Appendix C for an example of declaration screen). Overall, participants had to make a total of 6 reports.

Reports. On the first day of the experiment, we provided instructions to download and use the app. As the app can retrieve the data transmission history stored in the device before installation, in Report 0 participants reported their weekly consumption levels over the previous three weeks (*Pre-Experimental periods*, weeks -3, -2, and -1). Because the participants cannot change their past consumption in response to the beginning of the experiment, this strategy allows us to obtain unbiased measure of baseline data consumption. Starting from the second data declaration (Report 1), the participants were asked to report their consumption levels of both Wi-Fi and mobile data for only the previous week. For example, in Report 1 of the experiment (October 24), all participants were asked to report the consumption information from the week October 17-23. In reports 1 to 3, participants declared their consumption levels over *Experimental periods* (weeks 1, 2, and 3), while in reports 4 and 5 they declared consumption levels over *Post-Experimental periods* (weeks 4 and 5).

Incentives and data accuracy. Participants were paid €6 for full completion of the study, requiring them to miss at maximum one data declaration. Since we relied on participants to report their data consumption, incentive compatibility was ensured with a lottery. Subjects were instructed that in addition to the fixed amount earned for participation, truthfully reporting their actual consumption levels would make them eligible to win a prize of 200€. Specifically, at the end of the study, 20 participants per experimental condition (about 10% of the sample) were randomly selected and asked to send screenshots of the app showing actual Wi-Fi and mobile consumption levels for all weeks. Selected participants were awarded the lottery prize only if the information was consistent with that reported over the course of the experiment. Among the randomly selected winners, we did not find any incoherence between declarations and actual consumption levels from the app.

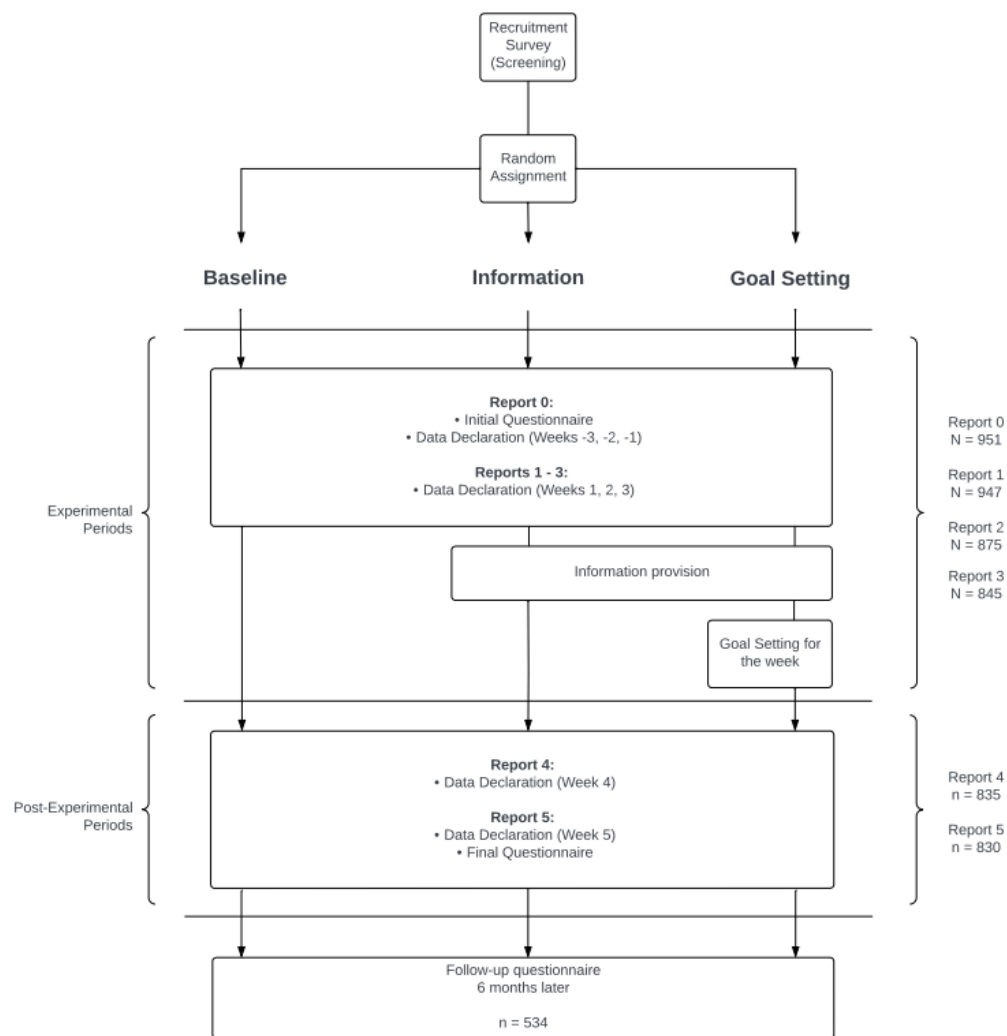
Experimental Conditions. Participants were randomly assigned to one of three experimental conditions. In the *Baseline* condition, participants did not receive any stimuli, and only had to declare their consumption levels. No additional information was given, nor questions asked, to participants in this condition during the six weeks. In the *Information* condition, participants received weekly information about their mobile consumption levels in the previous week, and the amount of CO₂ they would have saved via a Wi-Fi network (Appendix C). Lastly, in the *Goal setting* group, on top of receiving the same information as in the *Information* group, participants were asked to set a weekly goal regarding their level of Wi-Fi and mobile data consumption for the coming week (Appendix C). Participants could skip the goal page and set no goal. However, only 6 participants decided not to set goals over the entire reporting period. Goals for the week to come were made after declaring previous consumption levels.

The goal page reported the consumption from past week (in Gb) and participants could set any consumption level they desired. On the same webpage, they could use a calculator to compute the quantity of CO₂ their new consumption goal would produce in comparison to the previous week. Importantly, failure to fulfill a previous goal did not entail any consequences in the experiment. Each week participants were informed about their goal achievements by comparing goals with actual consumption levels (see Appendix C).

Our treatment provision in the *Information* and *Goal setting* conditions lasted three weeks (Weeks 1-3 in Figure 1). Afterwards (Weeks 4-5, Figure 1), participants were still required to declare their consumption levels but no longer received emissions information or requests for weekly goals.

Initial and Final Questionnaires. Reports 0 and 5 also included some complementary questionnaires (see Appendix D). In Report 0, we administered the *Initial Questionnaire* including demographics and individual characteristics, consumption modes (use frequency of domestic and public Wi-Fi networks), and smartphone settings (e.g., automatic app updates via Wi-Fi, auto-join of known Wi-Fi networks). We also assessed pre-experimental knowledge about CO₂ advantages of offloading. Items presented pairs of actions and asked participants to report their best guess as to which results in greater CO₂ production. As a benchmark from another domain, we included two analogous questions comparing transportation modes (e.g., flying vs. high-speed rails). These items were included only in Report 0, after data declaration, but before displaying the information and goal setting pages. We repeated a subset of the questions included in the *Initial Questionnaire* during the *Final Questionnaire* administered at the end of the experiment in Report 5 (see Appendix D).

Figure 1. Summary of the experimental design.



Notes: each Report included data declaration, and for experimental groups the provision of information (Appendix C) and goal setting opportunity (Appendix C). More detailed information about the experimental timeline is given in Table A1, panel a.

2.1 Sample

The participants were recruited from the subject pools of each laboratory and were invited using the recruitment platform ORSEE (Greiner, 2015). Upon acceptance to take part in the experiment, they were randomly assigned to one of the experimental conditions. Our initial sample on Report 0 was 951, of whom 830 completed the whole study. We find no evidence of attrition bias across conditions (Table A1 panel b). Our final sample is balanced in gender (54% female) and has an average age of 23 years old. All recorded demographic variables are comparable across experimental conditions (Table 1).³ Most participants have access to a domestic Wi-Fi network (85%), and nearly all have an unlimited mobile internet plan. While Wi-Fi infrastructure is widespread, only about half of participants report systematically using Wi-Fi when at home, and usage outside the home is limited. Finally, we exclude 21 observations (0.2% of the sample) reporting implausible data consumption values, likely due to unit misreporting (e.g., GB instead of MB).

Table 1. Summary Statistics and Balance tests.

	<i>Overall</i>	<i>Baseline</i>	<i>Information</i>	<i>Goal Setting</i>	<i>p</i>	<i>SMD</i>
<i>Female (%)</i>	448 (54.1)	137 (50.9)	153 (55.8)	158 (55.4)	0.443	0.066
<i>Age</i>	23.63 (5.05)	22.88 (3.66)	24.18 (5.43)	23.81 (5.69)	0.008	0.18
<i>Wi-Fi at home</i>	702 (84.8)	230 (85.5)	238 (86.9)	234 (82.1)	0.423	0.088
<i>Unlimited Mobile data plan (%)</i>	802 (96.9)	261 (97.0)	267 (97.4)	274 (96.1)	0.142	0.049
<i>Green Identity</i>	15.60 (3.04)	15.58 (3.19)	15.69 (2.96)	15.54 (2.99)	0.828	0.034
<i>Wi-Fi consumption (Gb) pre-experimental periods</i>	10.03 (17.43)	10.15 (17.99)	10.20 (11.48)	9.76 (21.27)	0.947	0.016
<i>Mobile consumption (Gb) pre-experimental periods</i>	7.10 (8.86)	7.42 (9.59)	6.59 (7.73)	7.30 (9.16)	0.495	0.064
<i>Location (N. participants)</i>						
<i>Dijon</i>	105 (12.7)	36 (13.4)	41 (15.0)	28 (9.8)		
<i>Montpellier</i>	162 (19.6)	50 (18.6)	54 (19.7)	58 (20.4)		
<i>Nice</i>	124 (15.0)	39 (14.5)	38 (13.9)	47 (16.5)		
<i>Strasbourg</i>	437 (52.8)	144 (53.5)	141 (51.5)	152 (53.3)		
<i>Total</i>	828	269	274	287	0.64	

Notes: *p*-value reports the results of ANOVA (for numerical variables) and Kruskal-Wallis (for categorical variables) tests, and for the number of participants across locations, of a Chi-square test (4x3 factors) is reported. The number of participants refers to those who participated until the end of the experiment (i.e., Week 5). Column SMD reports standardized mean differences.

³ We only report a small (1 year), yet significant, difference in age across conditions. SMD is lower than the conventional threshold of 0.20. Nevertheless, in all our analyses, we report results with and without demographic controls, including age.

3. Empirical Strategy

To estimate the impacts of the interventions econometrically, we use a difference-in-difference approach. The basic model we estimate is:

$$y_{it} = \alpha_i + Exp_{it} \cdot Treat_{it} \cdot \beta + Exp_{it} \cdot \gamma + Treat_{it} \cdot \delta + \mathbf{Z} \cdot \epsilon + e_{it} \quad (1)$$

where y represents either mobile data, Wi-Fi data or the percentage of Wi-Fi (i.e., offloading measure). We follow the literature in energy economics (Löschel et al., 2023) and logarithmize both mobile and Wi-Fi data consumption. This transformation allows us to interpret treatment effects across models in terms of percentage change.⁴ $Treat$ is a vector of dummy variables indicating our three experimental conditions with the *Baseline* as base category, Exp is an indicator turning on during experimental weeks (Weeks 1-3); β is the Average Treatment Effect (ATE) and the main parameter of our interest in what follows. For models using Mobile and Wi-Fi consumption, such a coefficient can be interpreted as the approximate average percentage change in consumption following our interventions. Coefficients α_i are random effects at the individual level. Finally, we control for a set of relevant variables ($\mathbf{Z}$) that include demographics, information on Wi-Fi and mobile plans and usage from the Initial Questionnaire, time preferences, personality, and green identity. Standard errors are clustered at the individual level to account for repeated observations.

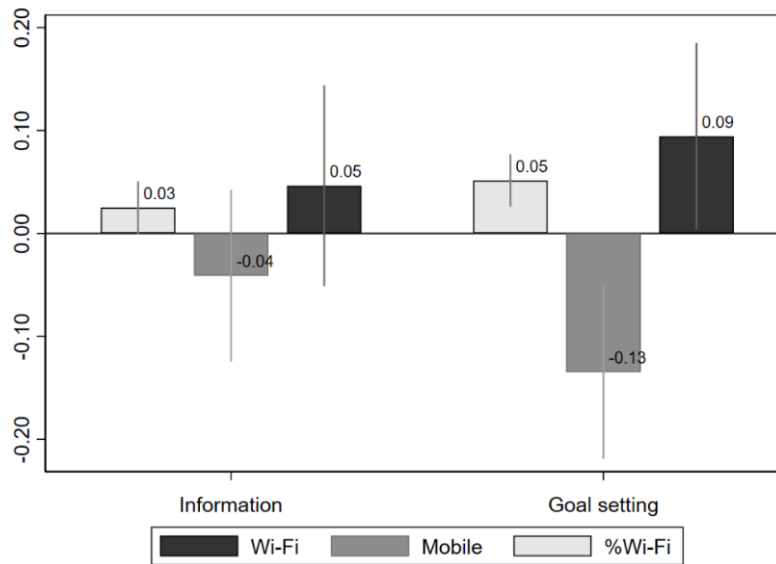
4. Main Results

Descriptive statistics for consumption levels are reported in Table A2-3 and Figure A2-3. Figure 2 and Table 2 report the ATEs from the difference-in-difference model. We also report standard deviation effect sizes using Glass's Δ , by dividing each treatment coefficient by the between-participant standard deviation of the outcome in the *Baseline* condition. Pre-experimental levels are similar across all conditions and measured variables indicating good baseline balance (i.e., treatment indicators, *Goal setting* and *Information*, are not significant; Table 2). ATE estimates show a significant treatment effect of the goal setting intervention of about 10% (0.11 s.d.) in Wi-Fi data use relative to the *Baseline* (columns 1-2). Our information intervention did not significantly change Wi-Fi data consumption. Consumption of mobile data significantly decreases in the *Goal setting* group relative to the *Baseline* of about 13.5% (0.14 s.d.), while we do not report any significant change in the *Information* condition (columns 3-4). Total amounts of data consumption (columns 5-6) show no overall difference relative to the change in the *Baseline* across any experimental condition. Consequently, the percentages of Wi-Fi increase significantly only in the *Goal setting* condition by about 5 percentage points, i.e., 0.18 s.d. (columns 7-8). All results remain robust even after including our control covariates.⁵

⁴ The logarithmization reduces the consequences of skewed data common of all variables expressing users' consumption levels. As a robustness check, in the Appendix, we also report models using non-logarithmized data (Table A5, models 7-10). Additionally, we report the results of Tobit models of mobile and Wi-Fi data, since all these variables are left-censored to 0 (Table A5, models 3-6). Our results remain unchanged compared to the models that we report in the main text.

⁵ See Table A4. Results hold unchanged to estimating linear regression specification instead of a Tobit model (Table A5, models 1-2).

Figure 2. Average Treatment Effects (ATEs) across experimental conditions and measures of data consumption.



Notes: depicted ATEs are expressed in percentage change and estimated from models in Table 2. Error bars report estimates of the 95% confidence interval. Reported figures rounded to the second decimal.

We report similar results when breaking down the treatment effect by week (Figure A4; Table A6). Percentages of Wi-Fi are significantly higher in *Goal setting* than those reported in *Baseline* over all experimental periods (from week 1 to 3), while they are not statistically different in the *Information* group.

Result 1. *Providing participants with technical information on consumption consequences alone produces no significant effect on data consumption levels, while combining information with goal setting significantly increases offloading relative to the Baseline.*

To assess whether goal setting produces a significant marginal effect relative to sheer information provision, we test whether the estimated treatment effect in the *Goal setting* condition differs from that in the *Information* condition. We report the results from linear combination tests on coefficients reported in Table 2. Full regression results using *Information* as the baseline category are in Table A7. For Wi-Fi data consumption is higher in *Goal Setting* by 4.7%, though this is not statistically significant ($p=0.343$). In contrast, mobile data consumption in *Goal Setting* is lower than that observed in *Information* by 9.3%, and statistically significant ($p=0.026$), suggesting that goal setting significantly reduces mobile data usage beyond what information alone achieves. Consequently, offloading rates in the *Goal setting* condition are statistically higher by 2.6 percentage points relative to the *Information* condition ($p=0.041$). Put together, the reduction in mobile data and the significant shift in the Wi-Fi share over the total leads us to conclude that:

Result 2. *Goal setting generates a significant behavioral response relative to information provision alone, primarily by reducing mobile data consumption and increasing offloading.*

Table 2. Regression estimates of consumption levels.

	Log(Wi-Fi)		Log(Mobile)		Log(Total)		% Wi-Fi	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Exp	0.0978** (0.0320)	0.0264 (0.0443)	-0.0150 (0.0308)	0.0816 (0.0417)	0.0858** (0.0313)	0.0632 (0.0376)	0.0270** (0.00932)	-0.0131 (0.0160)
Exp × Goal setting	0.101* (0.0465)	0.0943* (0.0463)	-0.135** (0.0429)	-0.135** (0.0429)	-0.0765 (0.0418)	-0.0605 (0.0416)	0.0531*** (0.0130)	0.0514*** (0.0130)
Exp × Information	0.0462 (0.0497)	0.0464 (0.0498)	-0.0412 (0.0425)	-0.0412 (0.0425)	-0.0232 (0.0435)	-0.0235 (0.0435)	0.0256 (0.0131)	0.0251 (0.0131)
Goal setting	-0.110 (0.0815)	-0.0866 (0.0705)	-0.0381 (0.0797)	-0.0159 (0.0646)	-0.0347 (0.0649)	-0.0239 (0.0636)	-0.0124 (0.0289)	-0.0170 (0.0221)
Information	0.0457 (0.0851)	-0.0141 (0.0757)	-0.0712 (0.0776)	0.0358 (0.0646)	0.0139 (0.0669)	0.0503 (0.0643)	0.0136 (0.0293)	-0.0280 (0.0222)
Baseline	1.856*** (0.0584)	0.128 (0.375)	1.644*** (0.0563)	2.560*** (0.314)	2.586*** (0.0478)	2.434*** (0.353)	0.559*** (0.0208)	0.0726 (0.109)
Controls	N	Y	N	Y	N	Y	N	Y
Observations	4920	4908	4920	4908	6607	4941	4920	4908
Adj. R ² / AIC	0.007	0.213	0.005	0.304	0.003	0.085	411.782	-88.418

Notes: the table shows the results of a panel difference-in-difference regression specification with individual random effects and standard errors clustered at the individual level in parentheses. Coefficients on Exp × Information and Exp × Goal setting are the difference-in-differences estimates relative to Baseline. Models 1-7 are linear regressions and Models 7-8 use a Tobit regression. For models 7-8 AIC is reported. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Post Experimental periods and effect persistence. Our design allows us to explore the persistence of our interventions. In the last two weeks of the experiment (weeks 4 and 5), we removed both our *Information* and *Goal Setting* treatments so that participants neither received information, nor set new goals during this period. We compare our post-experimental periods to both experimental (weeks 1-3) and pre-experimental periods (weeks -1 to -3) to investigate the persistence of our intervention effects. If the treatment effect of our goal setting intervention persists also after it ended, we should expect no decay relative to experimental periods, and a significant difference relative to pre-experimental periods. Results from our estimates in Table A8 confirm this persistence. The percentage of Wi-Fi consumption remains unchanged between experimental and post-experimental periods in all experimental conditions (models 1-2). All interaction terms between treatment indicators and the indicator for post-experimental periods (*Post*) are insignificantly different from zero. In particular, the overall effect of our goal setting intervention did not decay upon its end. Similarly, when comparing post-experimental periods with pre-experimental ones, consumption levels in the former periods continue to be significantly higher than pre-experimental ones in the *Goal setting* condition only (models 3-4).

Effect size. How large is the effect of our goal setting intervention? By conventional benchmarks (e.g., Cohen’s standards) it is modest: the increase in the Wi-Fi share corresponds to roughly 0.18 standard deviations of the outcome in the *Baseline* condition. These benchmarks, however, are a poor guide for interventions deployed in the field and should be compared to average effect sizes from similar studies. For example, DellaVigna & Linos (2022) find that, across 126 randomized trials run by two large nudge units and covering 23 million individuals, the average intervention raises the targeted behavior by 1.4 percentage points, an 8.0% increase relative to the control group, while the average effect reported in academic journals is roughly six times larger — a gap attributable largely to publication bias. Our effect is of comparable relative magnitude to what such units achieve at scale, and well below the inflated academic average. Two features, however, distinguish it from the typical nudge. First, it is close to costless to deploy: a single screen in an application the firm already operates. Second, and more importantly, it does not decay immediately after the intervention stops — a property that most information-based interventions lack (Andor et al., 2022; Brandon et al., 2017), and one that matters when aiming to reduce greenhouse gas emissions.

4.1. Robustness check: *Pure control* condition

To rule out any possible behavioral change due to self-monitoring or Hawthorne effect in the *Baseline*, we recruited a fourth group of participants after the end of the main experiment. We asked participants in this condition (*Pure Control*) to download the monitoring app and declare *ex-post* the consumption levels over the same weeks as in our other three conditions. Put differently, the main difference between the *Baseline* and *Pure Control* conditions is that participants in the latter group declare all their data consumptions over the same experimental period at once, with no possibility to correct their consumption levels over time. In this way, any difference emerging in consumption levels relative to the *Baseline* condition can be attributed to self-monitoring or the effect of having continuous feedback. In total, 242 subjects were recruited and asked to declare the consumption over the same 8 weeks as in the other experimental conditions.⁶ We also administered the *Initial questionnaire* as in the other three conditions to collect data on participants’ characteristics and data usage profiles. None of our collected measures in the *Pure Control* condition are statistically different from those observed in the *Baseline* conditions (Table A9). By using the same models as in Eq. 1, results show no overall significant

⁶ This was possible as the application *GlassWire* used in this study allows to retrieve the history of data consumption up to 16 weeks prior installation.

differences in any data consumption measures between the *Baseline* and *Pure Control* condition (Table A10). Lastly, following our pre-registration, we also test whether there is a change when transitioning from pre- to experimental periods due to self-monitoring alone (see Table A11). Results show no significant change across periods. We therefore conclude that the changes observed in *Baseline* condition were due largely to natural “business cycles” of smartphone internet use.

5. Mechanisms Investigation: The role of attention and information recall

Attention. Providing information does not by itself imply that it will be retained. The target must also pay attention to learn it (Abrahamse & Matthies, 2018; Gao & Tavoni, 2023). To explore whether attention and saliency of information played a role in our context, we analyze data on the time that participants spent interacting with the information page. The underlying assumption is that the more time participants spend on this page, the more likely they are to integrate the information provided. Hence, the time spent on the information page is a proxy of how likely information acquisition is which, in principle, should lead to behavioral change. Notice that this is possible because participants received consumption information on a different screen from that in which they set their goals. Had the participants completed both tasks on the same page, the time spent would not represent a clean measure of attention, since it would also include the time required to make the goal decision.

Figure 6 shows the time spent on the information page, by conditions and experimental week. While participants spent on average 42 seconds on the page the first time they accessed it on week 1, participants in the *Information* condition significantly reduced the time spent starting from the second week (relative to week 1: -27 seconds in week 2; -31 seconds in week 3; t-test, all $p < 0.001$). In the *Goal setting* condition, data on time spent on the information page suggests the presence of different dynamics. After observing a slight drop in seconds spent on the information page from the first to the second experimental week (average difference between week 2 and week 1: -9 seconds, t-test, $p = 0.036$), time spent remains stable over the remaining weeks. On average, the time spent on the information page after the first week is 30 seconds in the *Goal setting* condition, an amount twice that of the average among participants in the *Information* group. This evidence suggests that the saliency of the information provided in our *Information* treatment was probably low, and that individuals did not attribute a high instrumental value to it. Coherently with our main results on data consumption, higher levels of mobile consumption are associated with lower attention to information in the past week (Table A12).

Beyond these dynamic patterns, we also report some insights on the relation between attention and a set of individual characteristics (Table A13). Individuals with stronger environmental concern spend more time on the feedback pages. These findings are consistent with theories of rational inattention: when the private utility of information is higher, individuals allocate more attention to it, whereas information perceived as redundant is more easily ignored (Golman et al., 2017; Sallee, 2014; Sims, 2010). Individual preferences (e.g., time preferences and stated Wi-Fi vs mobile preferences) are not associated with time spent on the information page, and controlling for them leaves the Information-condition decay in time spent intact.

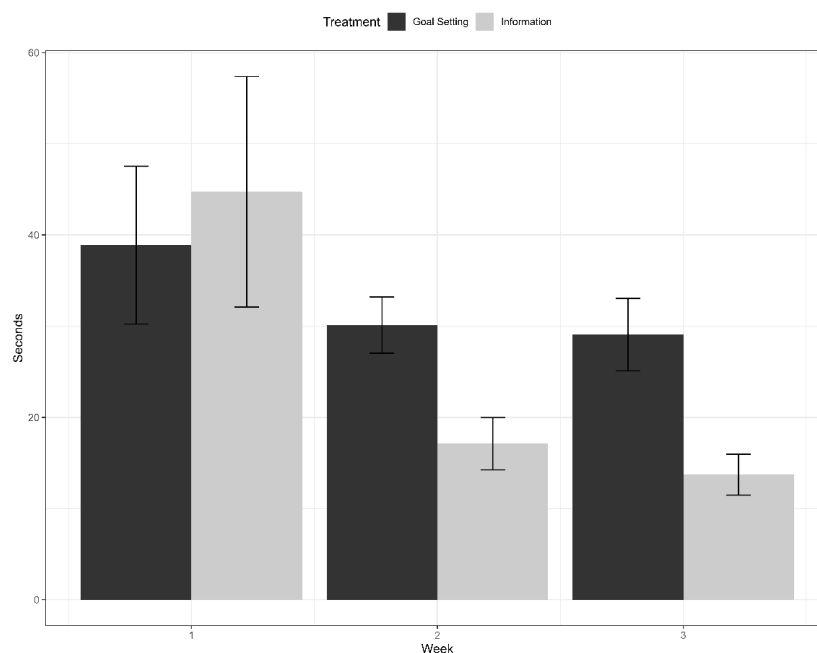
Follow-up Study and Information Recall. Has the information provision filled the knowledge gap previously documented in our pre-experimental survey? Six months after the experiment, we run a pre-registered⁷ follow-up study which included the same items as in our pre-experimental survey about knowledge of the CO₂ savings due to offloading and in other unrelated domains. A total of 534 participants took part in our follow up questionnaire, which represents 64% of the final sample included in the experiment. Attrition rates were similar across conditions (Table A1 panel b).

⁷ Pre-registration can be found at https://aspredicted.org/1ZB_FHF. Further details are in Appendix G.

Results show that correct answers from participants assigned to the *Goal setting* condition significantly improved (12.56% increase; proportion test on the number of correct answers, $p = 0.017$), while participants in the *Information* condition did not (3.49% increase; proportion test, $p = 0.57$). As expected, correct answers in the *Baseline* were also statistically equivalent to our pre-experimental levels (-3.35%, proportion test, $p = 0.59$). Table 3 shows complementary results using a difference-in-difference approach. We also find that participants who spent more time on the information page are more likely to recall the information correctly six months later (Table A14). As further evidence in support of these findings, we do not find any significant change across conditions when analyzing the other two items assessing knowledge about pollution related to transports, which were not the objects of our interventions (Table 3). In other words, participants in *Goal setting* spent more time on information pages, a proxy of their engagement with the provided information, and displayed better information recall over time, whereas no significant improvement is observed in the *Information* or *Baseline* conditions.

Other mechanisms. Finally, we rule out an alternative explanation relying on technological factors, rather than behavioral ones. The marked behavioral change among participants in the *Goal setting* group may be the consequence of some smartphone features that help in automated data management. Examples of such technological help are automatic join of known Wi-Fi networks (an option available on all smartphones), or the option of downloading apps and smartphones updates by Wi-Fi only. To address this potential channel, in both our initial and final questionnaire, we asked participants' information about automatic join and predefined network for updates. Confronting these measures over pre- and post-experimental periods show no significant differences (Table A15). Before the start of the experiment, around 41% of participants on average across all conditions declared to have automatic join of known Wi-Fi networks on their smartphone, while 53% on average use Wi-Fi networks for important smartphone updates. These percentages are similar across conditions, and do not significantly change upon our treatment provision. Similarly, we do not find any significant difference in the frequency of hotspot use across pre- and post-experimental periods.

Figure 6. Time (in seconds) spent on average on the information page, across treatments and experimental weeks.



Notes: error bars report the 95% confidence interval of the mean.

Table 3. Difference-in-Difference specifications of knowledge items.

	Offloading (1)	Offloading (2)	High-Speed trains (3)	Regional trains (4)
Dep. Variable: <i>Correct Answer</i>	OLS	Logit	OLS	OLS
Goal setting	-0.205 (0.114)	-0.875 (0.343)	0.094 (0.076)	0.058 (0.055)
Information	-0.061 (0.116)	-0.256 (0.492)	0.161* (0.077)	0.075 (0.056)
Follow-up	-0.034 (0.051)	-0.140 (0.216)	0.128*** (0.034)	0.061* (0.025)
Follow-up * Goal setting	0.159* (0.072)	0.683* (0.308)	-0.063 (0.048)	-0.034 (0.035)
Follow-up * Information	0.067 (0.073)	0.291 (0.312)	-0.094 (0.049)	-0.056 (0.036)
Constant	0.648*** (0.081)	0.606 (0.343)	0.682*** (0.054)	0.849*** (0.039)
Observations	1068	1068	1068	1068

Notes: The dependent variable is an indicator turning on if the answer given by the participant to the knowledge item is correct. *Exp* is an indicator turning on when the answer comes from pre-experimental periods (before the start of the experiment) and taking value zero for observations collected in our follow-up questionnaire (after 6 months). Standard errors clustered at the individual level in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

5. Conclusions and Implications

Organizations are increasingly accountable for the emissions their customers generate in using their products, yet the instrument they most often reach for — information provision — tends to raise awareness without changing behavior. This paper has shown that a simple, low-cost complement, self-set goals, can turn information provision into lasting behavioral change.

Common policy responses rely on information salience and filling knowledge gaps (Gao & Tavoni, 2023) by informing users about more sustainable choices (Allcott & Knittel, 2019; Allcott & Mullainathan, 2010; Gosnell et al., 2020). These approaches rest on the assumption that individuals are willing to act but lack knowledge about how to do so. Yet the challenge of changing behavior extends beyond correcting knowledge gaps: attention is limited, and individuals must decide where to allocate it. In an environment saturated with competing messages, the ability to capture and sustain attention to valuable information becomes crucial for effective policy design.

In this work, we have addressed a policy-relevant issue by using goal setting as a way to increase the instrumental value of information. The literature at the intersection of economics and psychology recognizes attention as goal-directed, guided by the decision-maker's objectives and motivations (Bordalo et al., 2022; Desimone & Duncan, 1995; Dijksterhuis & Aarts, 2010; Sallee, 2014; Sims, 2010). In economics, models of rational inattention (Sallee, 2014; Sims, 2010) and sparsity (Gabaix, 2014) formalize this selective allocation as an optimization problem in which individuals process information only when its expected value exceeds the cognitive cost. By prompting individuals to define explicit goals, goal setting can shift these attention weights, increasing the perceived value of relevant information.

Results from our field experiment show that information provision alone is not enough to bring about significant behavioral change. At the same time, the simple addition of self-set consumption goals shifted data consumption levels. This intervention prompted a behavioral shift in the type of network used without altering the overall quantity of data consumed, pointing to a substitution rather than a conservation effect. The effect persists up to two weeks after the removal of our stimuli. The evidence of decaying attention and lower information recall in the *Information* condition relative to *Goal setting* suggests that goal setting made information salient in the short term, while helping address the knowledge gap in the long term. These results are in line with the evidence of sustained behavioral change in the *Goal setting* group during post-experimental periods.

Put together, our results caution against the intuitive managerial response of simply sending users more information: attention to repeated messages decayed sharply in our *Information* condition, and it was the goal, not additional messaging, that kept the same information salient. While our study demonstrates a cost-effective behavioral strategy to encourage users to adopt a pro-environmental digital behavior, it is reasonable to question whether technological solutions might be more effective (Brandon et al., 2017). For instance, regulators could simply require smartphone producers that Wi-Fi offloading be activated by default. Such a technological solution could offer a more straightforward and scalable alternative. However, we find no evidence that participants changed their smartphone settings to enable the automatic use of Wi-Fi networks. These findings highlight that, even in the presence of technological shortcuts, setting goals can promote behavioral change through attention and memory.

Our results have several implications. On the theoretical side, our findings provide evidence of the importance of attention and memory in shaping consumption decisions, reinforcing recent calls to integrate these mechanisms more explicitly into models of behavior (Graeber et al., 2024). On the practical side, our findings speak to organizations — a telecommunications operator in our case, but more broadly any firm that reaches users through a digital interface — that are increasingly accountable for the emissions their customers generate in using their products (Hertwich & Wood, 2018). Self-set goals are a low-cost and scalable way to act on this responsibility: a single screen in an application was enough to sustain the attention users paid to emissions information and to shift their behavior durably. Two features make the lever attractive to managers. First, it changed the composition of data use toward Wi-Fi without reducing total consumption, so the firm lowers the use-phase emissions it must report without asking customers to give up the service they value. Second, because the effect persisted after the intervention ended, the intervention does not require continuous prompting to remain effective. Concretely, if the estimated effects in the Goal setting condition persisted, they would imply a reduction in CO₂ emissions of 2.4 kg per person per year — equivalent to 167 days of a LED bulb on, 300 km in a high-speed train, or 44 hours of electric heating.⁸ The magnitude is modest in absolute terms but compares favorably with behavioral interventions implemented at scale (e.g., 1.4 percentage points, an

⁸ Estimates based on the IByte Model from “The Shift Project” (available at: <https://theshiftproject.org/wp-content/uploads/2018/10/Lean-ICT-Materials-1byte-Model-2018.xlsx>) to estimate the CO₂ emissions potentially saved. Our estimates assume that the effect reported in our goal setting intervention does not decay over long periods of time, and that people do not habituate to the intervention.

8.0% increase on average; Della Vigna & Linos, 2022). Our five percentage-point shift is of comparable relative size and, unlike most such interventions, persists after the intervention ends.

We hope this work encourages further research, especially using field experiments, to examine the behavioral and cognitive mechanisms that drive real-world behaviors and evaluate organizational environmental policies.

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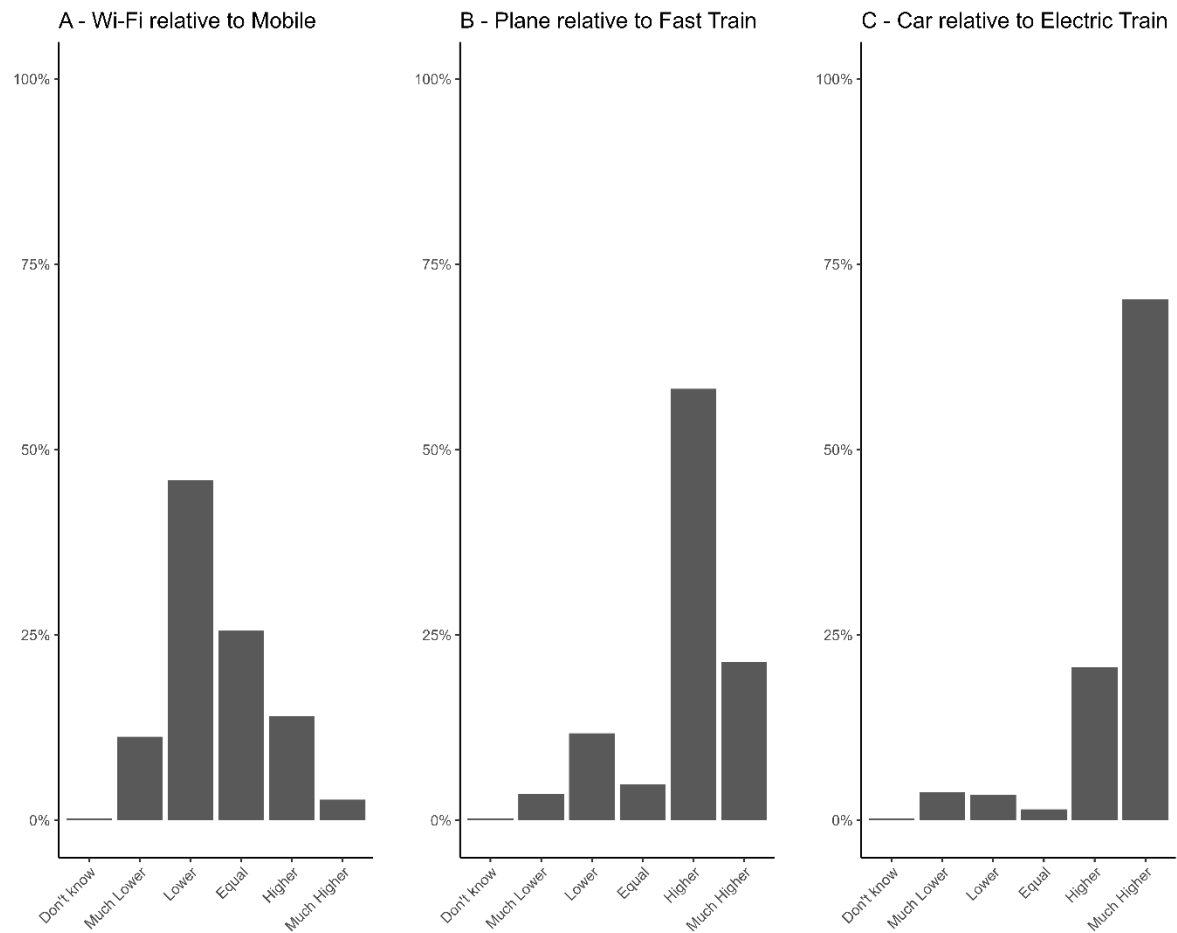
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Online Appendix

- A. Additional Figures**
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A. Additional Figures

Figure A1. Percentage of respondents by experimental condition, over the whole experimental timeline.



Notes: Panel A shows the results to the question “In your opinion, consuming 100 Gb of internet on a smartphone using the Wi-Fi network compared to consuming the same amount of Gb on a mobile network (4G/5G) produces an amount of CO2...”. Panel B shows the responses to “In your opinion, travelling 10 km by plane compared to travelling 10 km by TGV (fast train) produces a quantity of CO2...”. Lastly, Panel C reports the responses to “In your opinion, travelling 10 km by car (internal combustion engine) compared to travelling 10 km by TER (electric train) produces a quantity of CO2...”. These questions were part of the *Initial Questionnaire* (see Appendix D, *Knowledge about offloading* section).

Figure A2. Percentage of Wi-Fi by experimental condition, over the whole experimental timeline.

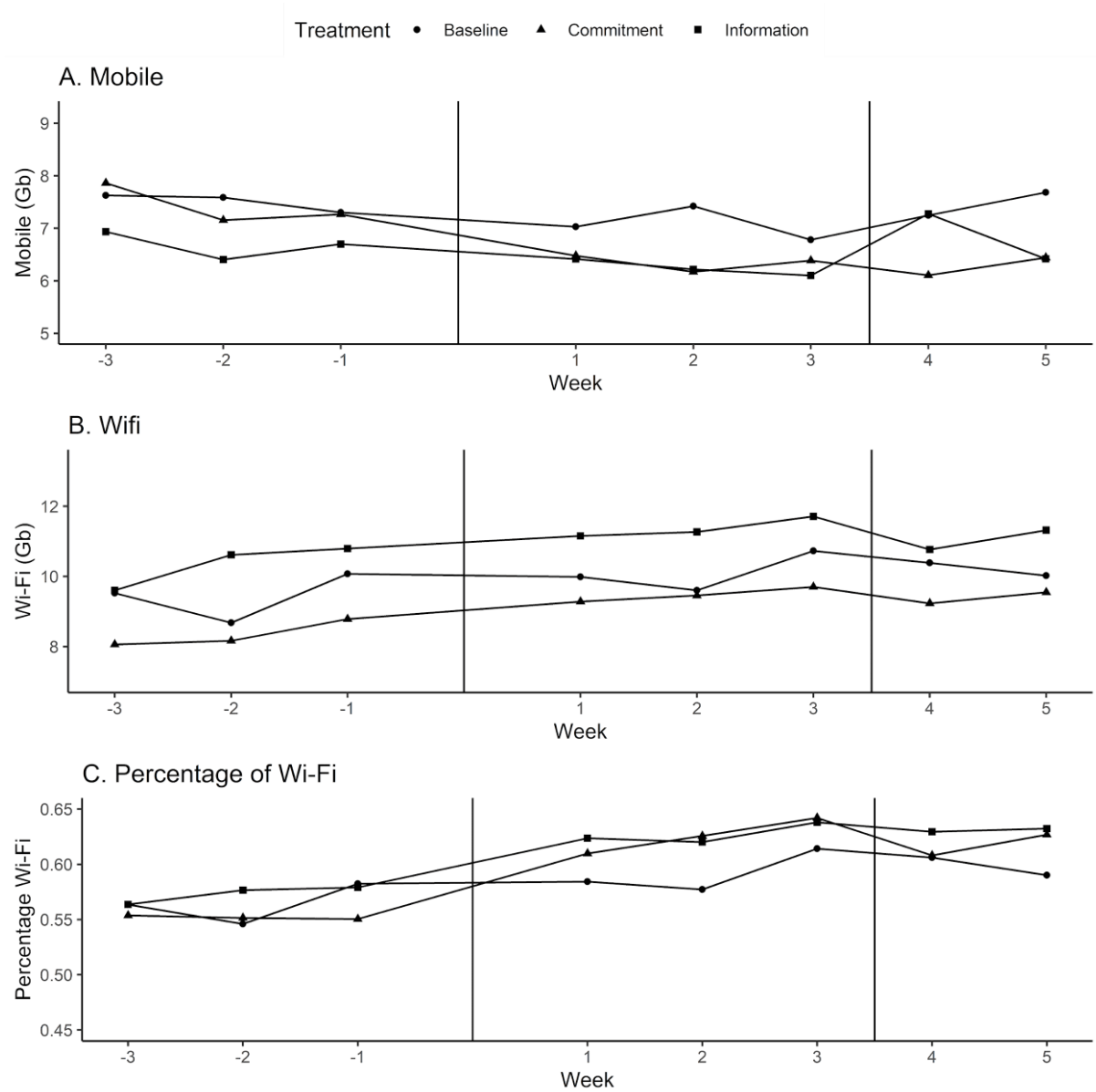


Figure A3. Consumptions (Gb) and %Wi-Fi by experimental condition, Before, During and After each intervention.

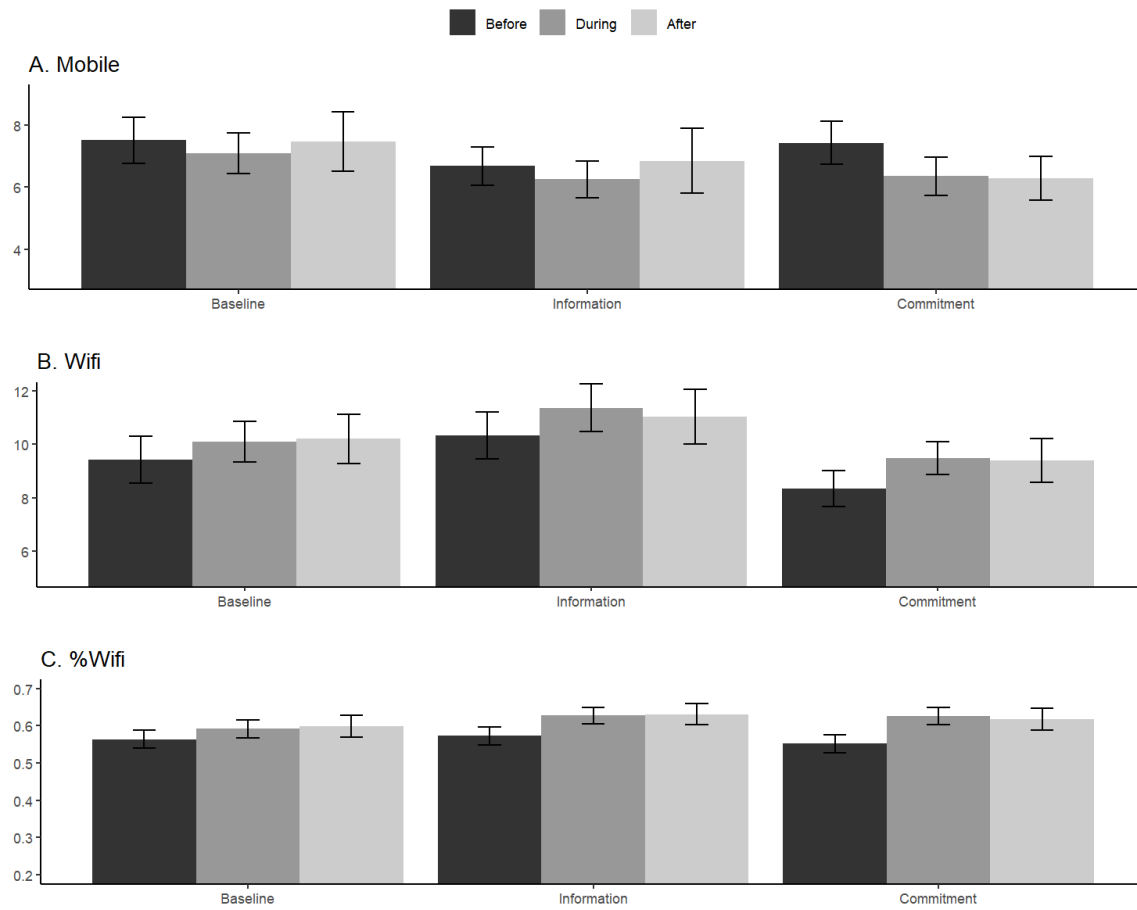
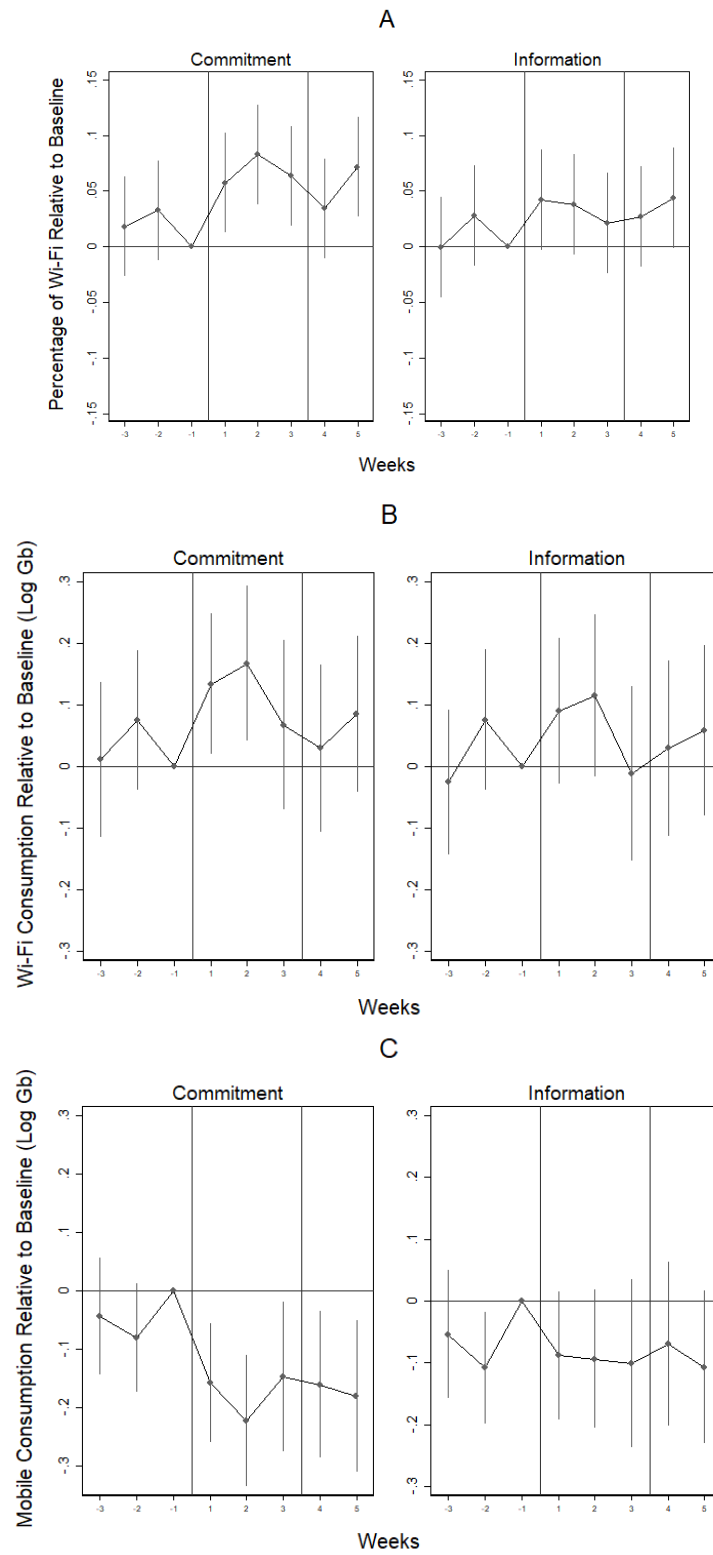


Figure A4. Average treatment effects in *Goal setting* and *Information* conditions relative to *Baseline*.



Notes: Estimates from the corresponding model are reported in the Appendix (see Table A6). Panel A reports the change in % Wi-Fi; Panel B reports the change in Wi-Fi data (Log Gb); Panel C reports the change in Mobile data (Log Gb). Error bars depict the 95% confidence interval.

B. Additional Tables

Table A1. Experimental timeline (panel a) and attrition analysis (panel b).

<i>Panel a – Experimental timeline</i>			
Phase	Date	Reported Weeks (Consumption data)	Sample size
Recruitment and Report 0	24 Oct.	Week -3 (Oct. 3 – Oct. 9) Week -2 (Oct. 10-16) Week -1 (Oct. 17-23)	2310 contacted via laboratory recruitment platforms 287 excluded (not meeting criteria) 502 not replying screening question 951 completed all tasks across conditions: • 514 – <i>Baseline</i> • 490 – <i>Information</i> • 507 – <i>Goal setting</i>
Report 1	31 Oct.	Week 1 (Oct. 24-30)	947 completed
Report 2	7 Oct.	Week 2 (Oct. 31 – Nov. 6)	875 completed
Report 3	14 Nov.	Week 3 (Nov. 7-13)	845 completed
Report 4	21 Nov.	Week 4 (Nov. 14-20)	835 completed
Report 5	28 Nov.	Week 5 (Nov. 21-27)	830 completed • 269 – Baseline • 274 – Information • 287 – Goal setting
<i>Panel b – Attrition analysis of Main Study and Follow-up study (observations at the individual level)</i>			
		<i>Dependent variable:</i>	
		Wave 1	Wave 2
		Participant dropped the study	Participant did not follow up
Goal Setting		-0.123 (0.112)	0.122 (0.178)
Information		-0.034 (0.112)	0.165 (0.180)
Constant		0.377*** (0.079)	-0.688*** (0.129)
Observations		1,978	830

Notes: Panel b reports logit regression at the individual level. Dependent variables are indicators taking value of 1 if the participant has dropped the study (Wave 1) or has not replied to join the second wave. *p<0.05, **p<0.01, ***p<0.001

Table A2. Consumption levels

Treatment		Mobile (Gb)	Wi-Fi (Gb)	% Wi-Fi
<i>Baseline</i>	Before	7.50	9.43	0.56
	During	7.08	10.11	0.59
	After	7.46	10.21	0.60
<i>Information</i>	Before	6.68	10.34	0.57
	During	6.24	11.38	0.63
	After	6.85	11.04	0.63
<i>Goal setting</i>	Before	7.43	8.34	0.55
	During	6.35	9.48	0.63
	After	6.27	9.39	0.62

Table A3. Summary Statistics: change in consumption of data and %Wi-Fi between pre-experimental (Weeks -3, -2, -1) and experimental periods (Weeks 1, 2, 3).

Condition	$\Delta Mobile$	$\Delta WiFi$	$\Delta \% WiFi$
Baseline	-0.37	0.69	2.43%
Information	-0.31	1.09	5.28%
Goal setting	-1.09	1.20	7.44%

Notes: figures in the second and third columns report raw changes in data consumption (in Gb) of Mobile and Wi-Fi. The fourth column reports the percentage change of Wi-Fi over the total (Mobile + Wi-Fi).

Table A4. Regression models of consumption measures. Baseline condition is *Baseline*.

	(1) Log(Wi-Fi)	(2) Log(Wi-Fi)	(3) Log(Mobile)	(4) Log(Mobile)	(5) Log(Total)	(6) Log(Total)	(7) % Wi-Fi	(8) % Wi-Fi
Expt	0.0978** (0.0320)	0.0264 (0.0443)	-0.0150 (0.0308)	0.0816 (0.0417)	0.0858** (0.0313)	0.0632 (0.0376)	0.0270** (0.00932)	-0.0131 (0.0160)
Goal setting	-0.110 (0.0815)	-0.0866 (0.0705)	-0.0381 (0.0797)	-0.0159 (0.0646)	-0.0347 (0.0649)	-0.0239 (0.0636)	-0.0124 (0.0289)	-0.0170 (0.0221)
Information	0.0457 (0.0851)	-0.0141 (0.0757)	-0.0712 (0.0776)	0.0358 (0.0646)	0.0139 (0.0669)	0.0503 (0.0643)	0.0136 (0.0293)	-0.0280 (0.0222)
Expt * Goal setting	0.101* (0.0465)	0.0943* (0.0463)	-0.135** (0.0429)	-0.135** (0.0429)	-0.0765 (0.0418)	-0.0605 (0.0416)	0.0531*** (0.0130)	0.0514*** (0.0130)
Expt * Information	0.0462 (0.0497)	0.0464 (0.0498)	-0.0412 (0.0425)	-0.0412 (0.0425)	-0.0232 (0.0435)	-0.0235 (0.0435)	0.0256 (0.0131)	0.0251 (0.0131)
Week		0.0178 (0.00924)		-0.0242** (0.00800)		0.00469 (0.00645)		0.0101** (0.00324)
Age		-0.0157* (0.00623)		-0.0179*** (0.00434)		-0.0238*** (0.00588)		-0.00202 (0.00175)
Female		0.00239 (0.0553)		-0.207*** (0.0483)		-0.108* (0.0460)		0.0502** (0.0174)
Green Identity		-0.0204* (0.00955)		0.00794 (0.00867)		-0.00665 (0.00831)		-0.00492 (0.00300)
Time		0.00325		-0.0373*		-0.0201		0.00623

		(0.0166)		(0.0145)		(0.0137)		(0.00483)
Has Wi-Fi at home		1.089*** (0.0745)		-0.835*** (0.0760)		-0.0716 (0.0698)		0.424*** (0.0270)
Has Forfait		0.560* (0.220)		0.435** (0.151)		0.510* (0.219)		-0.108* (0.0505)
Wi-Fi external		0.0582* (0.0283)		-0.0974*** (0.0263)		-0.0260 (0.0258)		0.0439*** (0.00848)
Wi-Fi Automatic		0.136** (0.0459)		-0.0483 (0.0412)		0.0626 (0.0405)		0.0544*** (0.0145)
Wi-Fi updates		0.0901* (0.0409)		-0.0137 (0.0345)		0.0682 (0.0355)		0.0360** (0.0129)
Hotspot		-0.0511 (0.0288)		0.220*** (0.0279)		0.0839*** (0.0247)		-0.0660*** (0.00901)
Prefers Wi-Fi		0.229*** (0.0622)		-0.141** (0.0540)		0.0350 (0.0511)		0.101*** (0.0188)
Extraversion		0.0220 (0.0243)		0.00954 (0.0205)		0.0121 (0.0200)		0.000586 (0.00759)
Agreeableness		-0.0106 (0.0314)		-0.00185 (0.0272)		0.00447 (0.0265)		-0.00427 (0.00975)
Constant	1.856*** (0.0584)	0.128 (0.375)	1.644*** (0.0563)	2.560*** (0.314)	2.586*** (0.0478)	2.434*** (0.353)	0.559*** (0.0208)	0.0726 (0.109)
Observations	4920	4908	4920	4908	6607	4941	4920	4908

Notes: 1-7 linear regression models, 7-8 Tobit models. All models include random effects at the individual level. Standard errors clustered at the individual level in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A5. Linear regressions (models 1-2), Tobit (columns 3-6) and Poisson (columns 7-10) regression models.

	(1) % Wi-Fi	(2) % Wi-Fi	(3) Log(Wi-Fi)	(4) Log(Wi-Fi)	(5) Log(Mobile)	(6) Log(Mobile)	(7) Wi-Fi	(8) Wi-Fi	(9) Mobile	(10) Mobile
Expt	0.025** (0.009)	-0.012 (0.013)	0.102*** (0.029)	0.027 (0.049)	-0.016 (0.024)	0.085* (0.042)	0.068*** (0.016)	-0.068* (0.027)	-0.054** (0.019)	0.032 (0.033)
Goal setting	-0.013 (0.028)	-0.017 (0.021)	-0.112 (0.085)	-0.086 (0.074)	-0.044 (0.078)	-0.016 (0.064)	-0.132 (0.083)	-0.144 (0.077)	-0.003 (0.092)	-0.008 (0.081)
Information	0.008 (0.028)	-0.030 (0.022)	0.053 (0.085)	-0.011 (0.074)	-0.077 (0.079)	0.035 (0.065)	0.090 (0.084)	0.013 (0.079)	-0.121 (0.093)	0.026 (0.082)
Expt * Goal setting	0.050*** (0.014)	0.048*** (0.014)	0.106** (0.040)	0.100* (0.040)	-0.137*** (0.034)	-0.137*** (0.034)	0.067** (0.023)	0.062** (0.023)	-0.110*** (0.026)	-0.109*** (0.026)
Expt * Information	0.027 (0.015)	0.026 (0.015)	0.043 (0.040)	0.043 (0.040)	-0.041 (0.034)	-0.041 (0.034)	0.025 (0.022)	0.025 (0.022)	0.002 (0.027)	0.002 (0.027)
Week		0.009** (0.003)		0.019 (0.010)		-0.025** (0.009)		0.034*** (0.006)		-0.021** (0.007)
Age		-0.001 (0.002)		-0.019** (0.006)		-0.018*** (0.005)		-0.015* (0.006)		-0.028*** (0.007)
Female		0.048** (0.016)		0.000 (0.059)		-0.213*** (0.051)		-0.054 (0.063)		-0.237*** (0.067)

Green Identity	-0.005 (0.003)	-0.018 (0.010)	0.009 (0.009)	-0.030** (0.011)	-0.009 (0.011)
Time	0.008 (0.005)	0.001 (0.016)	-0.036* (0.014)	0.001 (0.018)	-0.027 (0.018)
Has Wi-Fi at home	0.389*** (0.023)	1.187*** (0.091)	-0.832*** (0.079)	1.236*** (0.098)	-0.866*** (0.099)
Has Forfait	-0.046 (0.048)	0.563*** (0.166)	0.610*** (0.148)	0.284 (0.179)	0.359 (0.195)
Wi-Fi external	0.037*** (0.008)	0.071* (0.029)	-0.102*** (0.025)	0.026 (0.031)	-0.131*** (0.030)
Wi-Fi Automatic	0.049*** (0.013)	0.149** (0.049)	-0.051 (0.042)	0.104 (0.054)	-0.055 (0.056)
Wi-Fi updates	0.027* (0.011)	0.101* (0.044)	-0.026 (0.038)	0.089 (0.048)	-0.032 (0.053)
Hotspot	-0.061*** (0.009)	-0.058 (0.030)	0.228*** (0.026)	-0.059 (0.033)	0.253*** (0.033)
Prefers Wi-Fi	0.093*** (0.018)	0.244*** (0.064)	-0.150** (0.055)	0.240*** (0.068)	-0.145* (0.071)
Extraversion	0.001	0.023	0.011	0.007	0.017

		(0.007)		(0.026)		(0.022)		(0.027)		(0.029)
Agreeableness		-0.006		-0.003		-0.002		0.036		0.005
		(0.009)		(0.033)		(0.029)		(0.035)		(0.037)
Constant	0.566***	0.075	1.824***	-0.063	1.633***	2.376***	2.245***	0.854*	2.011***	3.328***
	(0.019)	(0.098)	(0.061)	(0.364)	(0.056)	(0.317)	(0.060)	(0.389)	(0.066)	(0.420)
Observations	4920	4908	4920	4908	4920	4908	4920	4908	4920	4908
Adjusted R^2										

Notes: All models include random effects at the individual level. Standard errors clustered at the individual level in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A6. Regression model showing Average Treatment Effect by week.

	(1) Log(Wi-Fi)	(2) Log(Mobile)	(3) % Wi-Fi
Week -3	-0.054 (0.045)	0.036 (0.033)	-0.018 (0.016)
Week -2	-0.103* (0.043)	0.057 (0.034)	-0.032 (0.016)
Week 1	0.036 (0.040)	0.043 (0.037)	0.001 (0.016)
Week 2	-0.001 (0.044)	0.055 (0.042)	-0.003 (0.016)
Week 3	0.104* (0.050)	-0.050 (0.049)	0.034* (0.016)
Week 4	0.060 (0.048)	-0.001 (0.046)	0.026 (0.016)
Week 5	0.053 (0.046)	0.016 (0.046)	0.009 (0.016)
Goal setting	-0.115 (0.079)	0.029 (0.068)	-0.034 (0.026)
Information	-0.027 (0.083)	0.091 (0.070)	-0.036 (0.026)
Week -3 # Goal setting	0.011 (0.064)	-0.044 (0.051)	0.018 (0.023)
Week – 3 # Information	-0.025	-0.054	-0.001

	(0.060)	(0.053)	(0.023)
Week – 2 # Goal setting	0.076 (0.058)	-0.082 (0.047)	0.033 (0.023)
Week – 2 # Information	0.075 (0.058)	-0.109* (0.046)	0.028 (0.023)
Week 1 # Goal setting	0.134* (0.058)	-0.158** (0.052)	0.057* (0.023)
Week 1 # Information	0.090 (0.060)	-0.088 (0.053)	0.042 (0.023)
Week 2 # Goal setting	0.167** (0.064)	-0.223*** (0.057)	0.083*** (0.023)
Week 2 # Information	0.115 (0.067)	-0.094 (0.057)	0.038 (0.023)
Week 3 # Goal setting	0.067 (0.070)	-0.147* (0.065)	0.064** (0.023)
Week 3 # Information	-0.011 (0.072)	-0.101 (0.070)	0.021 (0.023)
Week 4 # Goal setting	0.029 (0.069)	-0.161* (0.064)	0.034 (0.023)
Week 4 # Information	0.030 (0.073)	-0.070 (0.067)	0.027 (0.023)
Week 5 # Goal setting	0.085 (0.064)	-0.181** (0.066)	0.072** (0.023)
Week 5 # Information	0.059 (0.070)	-0.107 (0.063)	0.044 (0.023)

Age	-0.016** (0.006)	-0.018*** (0.004)	-0.002 (0.002)
Female	0.003 (0.055)	-0.215*** (0.047)	0.052** (0.017)
Green Identity	-0.018 (0.010)	0.003 (0.008)	-0.003 (0.003)
Time preferences	-0.002 (0.017)	-0.029* (0.014)	0.003 (0.005)
Wi-Fi at home	1.097*** (0.074)	-0.865*** (0.075)	0.430*** (0.027)
Mobile Plan Unlimited	0.507* (0.224)	0.471*** (0.143)	-0.129** (0.050)
Frequency Wi-Fi external	0.051 (0.028)	-0.089*** (0.026)	0.041*** (0.008)
Auto-Join Wi-Fi	0.123** (0.045)	-0.049 (0.041)	0.052*** (0.014)
Updates smartphone with Wi-Fi	0.092* (0.040)	-0.019 (0.034)	0.038** (0.013)
Frequency mobile hotspot	-0.048 (0.029)	0.218*** (0.027)	-0.065*** (0.009)
Prefers Wi-Fi over mobile	0.210*** (0.060)	-0.131* (0.053)	0.095*** (0.019)
Extraversion	0.027 (0.023)	0.011 (0.020)	0.002 (0.007)

Agreeableness	-0.011 (0.031)	0.003 (0.027)	-0.006 (0.010)
Intercept	0.248 (0.374)	2.555*** (0.306)	0.102 (0.107)
Observations	6546	6546	6546

Notes: All models include random effects at the individual level. Standard errors clustered at the individual level in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A7. Regression models of consumption measures. Baseline condition is *Information*.

	(1) Log(Wi-Fi)	(2) Log(Wi-Fi)	(3) Log(Mobile)	(4) Log(Mobile)	(5) % Wi-Fi	(6) % Wi-Fi
Post-Expe	0.144*** (0.038)	0.073 (0.050)	-0.056 (0.029)	0.040 (0.041)	0.053*** (0.009)	0.012 (0.016)
Baseline	-0.046 (0.085)	0.014 (0.076)	0.071 (0.078)	-0.036 (0.065)	-0.014 (0.029)	0.028 (0.022)
Goal setting	-0.156 (0.084)	-0.072 (0.072)	0.033 (0.078)	-0.052 (0.063)	-0.026 (0.029)	0.011 (0.022)
Post-Expe * Baseline	-0.046 (0.050)	-0.046 (0.050)	0.041 (0.042)	0.041 (0.043)	-0.026 (0.013)	-0.025 (0.013)
Post-Expe * Goal setting	0.055 (0.051)	0.048 (0.051)	-0.094* (0.042)	-0.094* (0.042)	0.028* (0.013)	0.026* (0.013)
Week		0.018 (0.009)		-0.024** (0.008)		0.010** (0.003)
Age		-0.016* (0.006)		-0.018*** (0.004)		-0.002 (0.002)
Female		0.002 (0.055)		-0.207*** (0.048)		0.050** (0.017)
Green Identity		-0.020* (0.010)		0.008 (0.009)		-0.005 (0.003)
Time		0.003 (0.017)		-0.037* (0.015)		0.006 (0.005)
Has Wi-Fi at home		1.089***		-0.835***		0.424***

		(0.074)		(0.076)		(0.027)
Has Forfait		0.560* (0.220)		0.435** (0.151)		-0.108* (0.051)
Wi-Fi external		0.058* (0.028)		-0.097*** (0.026)		0.044*** (0.008)
Wi-Fi Automatic		0.136** (0.046)		-0.048 (0.041)		0.054*** (0.014)
Wi-Fi updates		0.090* (0.041)		-0.014 (0.034)		0.036** (0.013)
Hotspot		-0.051 (0.029)		0.220*** (0.028)		-0.066*** (0.009)
Prefers Wi-Fi		0.229*** (0.062)		-0.141** (0.054)		0.101*** (0.019)
Extraversion		0.022 (0.024)		0.010 (0.020)		0.001 (0.008)
Agreeableness		-0.011 (0.031)		-0.002 (0.027)		-0.004 (0.010)
Constant	1.902*** (0.062)	0.114 (0.380)	1.573*** (0.053)	2.596*** (0.320)	0.573*** (0.021)	0.045 (0.110)
Observations	4920	4908	4920	4908	4920	4908

Notes: 1-4 linear regression models and 5-6 Tobit models. All models include random effects at the individual level. Standard errors clustered at the individual level in parentheses.

Table A8. Regression estimates of models comparing post-experimental periods to experimental and pre-experimental periods.

	(1)	(2)	(3)	(4)
	<i>Experimental vs. Post Experiment</i>		<i>Pre vs. Post Experiment</i>	
	% Wi-Fi	% Wi-Fi	% Wi-Fi	% Wi-Fi
Post Expt	0.005 (0.010)	-0.021 (0.014)	0.033** (0.010)	0.0001 (0.029)
Goal setting	0.037 (0.028)	0.032 (0.021)	-0.014 (0.028)	-0.018 (0.022)
Information	0.034 (0.028)	-0.004 (0.021)	0.011 (0.029)	-0.028 (0.022)
Post Expt * Goal setting	-0.015 (0.014)	-0.015 (0.014)	0.037* (0.014)	0.035* (0.014)
Post Expt * Information	0.002 (0.014)	0.002 (0.014)	0.028 (0.015)	0.028 (0.015)
Control variables	N	Y	N	Y
<i>N</i>	4107	4097	4097	4087

Notes: Tobit regression models with random effects at the individual level. Standard errors clustered at the individual level in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Variable *Post Expt* is an indicator turning on in periods in which the treatments were removed (weeks 4 and 5).

Table A9. Summary Statistics for *Pure Control* condition.

Treatment		Mobile (Gb)	Wi-Fi (Gb)	Offloading (%)
<i>Pure control</i>	Before	7.57	8.57	0.56
	During	7.12	8.90	0.57
	After	6.96	9.76	0.58

Table A10. Regression models comparing *Baseline* and *Pure Control*.

	(1) Log(Wi-Fi)	(2) Log(Wi-Fi)	(3) Log(Mobile)	(4) Log(Mobile)	(5) % Wi-Fi	(6) % Wi-Fi	(7) Log(Total)	(8) Log(Total)
Expt	0.021 (0.028)	-0.066 (0.051)	-0.022 (0.030)	0.017 (0.046)	0.006 (0.010)	-0.020 (0.019)	0.001 (0.021)	-0.057 (0.033)
Baseline	0.097 (0.088)	0.125 (0.080)	0.086 (0.086)	0.014 (0.073)	-0.014 (0.033)	0.020 (0.026)	0.089 (0.070)	0.069 (0.069)
Expt*Baseline	0.077 (0.042)	0.077 (0.043)	0.007 (0.043)	0.006 (0.043)	0.021 (0.014)	0.022 (0.014)	0.053 (0.034)	0.052 (0.034)
Week		0.022 (0.012)		-0.009 (0.010)		0.007 (0.004)		0.015 (0.008)
Age		-0.004 (0.006)		-0.012* (0.006)		0.001 (0.002)		-0.004 (0.006)
Female		0.030 (0.075)		-0.246*** (0.066)		0.077** (0.025)		-0.126* (0.063)
Green Identity		-0.012 (0.012)		0.020 (0.011)		-0.006 (0.004)		0.001 (0.010)
Time		-0.016 (0.024)		-0.010 (0.019)		0.000 (0.007)		-0.019 (0.017)
Has Wi-Fi at home		1.055*** (0.105)		-0.754*** (0.109)		0.413*** (0.040)		-0.038 (0.102)
Has Forfait		0.465		0.260		-0.082		0.442

		(0.245)		(0.193)		(0.062)		(0.235)
Wi-Fi external		0.066 (0.039)		-0.072* (0.036)		0.038** (0.012)		-0.000 (0.036)
Wi-Fi Automatic		0.092 (0.066)		-0.141* (0.063)		0.083*** (0.022)		-0.039 (0.065)
Wi-Fi updates		0.184*** (0.054)		0.033 (0.045)		0.047** (0.018)		0.142** (0.047)
Hotspot		-0.041 (0.041)		0.241*** (0.038)		-0.074*** (0.013)		0.148*** (0.037)
Prefers Wi-Fi		0.152 (0.087)		-0.190* (0.074)		0.097*** (0.027)		-0.026 (0.073)
Extraversion		0.000 (0.033)		0.019 (0.028)		-0.008 (0.011)		-0.016 (0.027)
Agreeableness		0.015 (0.042)		-0.002 (0.036)		-0.000 (0.014)		-0.003 (0.034)
Constant	1.759*** (0.066)	-0.218 (0.496)	1.558*** (0.065)	2.211*** (0.434)	0.573*** (0.024)	-0.022 (0.153)	2.520*** (0.053)	1.909*** (0.492)
Observations	2941	2941	2941	2941	2941	2941	2941	2941

Notes: All models include random effects at the individual level. Standard errors clustered at the individual level in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A11. Regression models comparing pre- and experimental periods in the *Baseline* condition.

	(1) % Wi-Fi	(2) % Wi-Fi	(3) Log(Mobile)	(4) Log(Mobile)
Expt	0.026** (0.009)	-0.023 (0.023)	-0.015 (0.031)	0.114 (0.058)
Week		0.012* (0.005)		-0.032* (0.014)
Age		-0.001 (0.004)		-0.020 (0.012)
Female		0.105*** (0.030)		-0.324*** (0.082)
Green Identity		-0.003 (0.005)		0.023 (0.014)
Time		0.009 (0.009)		-0.031 (0.023)
Has Wi-Fi at home		0.422*** (0.049)		-0.869*** (0.123)
Has Forfait		0.028 (0.088)		0.382 (0.245)
Wi-Fi external		0.029 (0.015)		-0.086 (0.047)
Wi-Fi Automatic		0.042		-0.040

		(0.026)		(0.078)
Wi-Fi updates		0.024 (0.022)		0.056 (0.055)
Hotspot		-0.068*** (0.016)		0.224*** (0.053)
Prefers Wi-Fi		0.084** (0.032)		-0.190* (0.091)
Extraversion		0.001 (0.013)		-0.001 (0.039)
Agreeableness		-0.003 (0.017)		0.001 (0.048)
Constant	0.555*** (0.020)	-0.025 (0.212)	1.644*** (0.056)	2.326*** (0.542)
Observations	1598	1598	1598	1598

Notes: 1-2 Tobit regressions, 3-4 linear regressions. All models include random errors at the individual level. Standard errors clustered at the individual level in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A12. Regression models of consumption variables on time spent on information page.

	(1)	(2)	(3)
	Log(Mobile)	Log(Wi-Fi)	% Wi-Fi
(Intercept)	2.23*** (0.32)	2.13*** (0.33)	0.48*** (0.11)
Log(Seconds _{t-1})	-0.11** (0.04)	-0.01 (0.04)	0.02^ (0.01)
Week	-0.01 (0.02)	-0.02 (0.02)	0.01 (0.01)
Controls	Y	Y	Y
<i>N</i>	1050	1050	1049

Notes: OLS regression models with standard errors clustered at the individual level in parentheses. Only experimental weeks and *Information* and *Goal setting* conditions considered. ^ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Controls include green identity and time preferences.

Table A13. OLS regression of the time spent by participants on the information page in each of the three experimental weeks over individual preferences and consumption levels.

	Time spent on Information Page (in seconds)
(Intercept)	23.09 *** (5.87)
Green Identity	1.03 * (0.48)
Wi-Fi at Home	-0.59 (4.18)
Prefers Wi-Fi	-0.81 (2.79)
Patience	0.41 (0.49)
Week 2	-8.85 (4.66)
Week 3	-10.55 * (4.79)
Information	5.96 (7.57)
Wi-Fi consumption (t-1)	-0.27 (0.16)
Mobile consumption (t-1)	0.23 (1.15)
Knowledge Mobile Carbon Emission Correct	-3.64 (2.86)
Week 2 * Information	-18.73 * (7.88)
Week 3 * Information	-21.30 ** (7.90)
R ²	0.06
Adj. R ²	0.05
Num. obs.	1050

Notes: Errors clustered at the individual level reported in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A14. Results from a Logit regression specification of correct recall in the Follow-up study about offloading knowledge.

	<i>Dependent variable:</i>
	Right answer (Follow-up study)
(Intercept)	0.332
Right answer in main study	0.012
Screen time	0.217*** (0.316)
Pseudo R ²	0.012
Observations	355

Notes: The dependent variable is a binary indicator equal to 1 if the participant correctly recalled in the follow-up study that mobile data pollutes more than Wi-Fi, and 0 otherwise. The main independent variable is the time spent reading the informational content during the study phase. The model controls for baseline knowledge, measured as whether the participant answered correctly in the main study. Estimates are obtained using a logistic regression model. Errors clustered at the experimental condition level. *p<0.05 **p<0.01 ***p<0.001

Table A15. Regression models of technological settings (auto join settings, frequency of hotspot using mobile network, and system updates using Wi-Fi)

	<i>Dependent variable:</i>		
	(1) Wi-Fi autojoin	(2) Mobile Hotspot	(3) Updates via Wi-Fi
Goal setting	0.051 (0.053)	-0.117 (0.118)	0.098* (0.052)
Information	0.155*** (0.054)	-0.157 (0.118)	0.050 (0.052)
Expt	0.013 (0.009)	-0.014 (0.020)	0.018** (0.009)
Goal setting * Expt	0.007 (0.012)	0.011 (0.027)	-0.005 (0.012)
Information * Expt	-0.015 (0.013)	-0.001 (0.028)	0.008 (0.012)
Constant	0.378*** (0.038)	2.658*** (0.084)	0.539*** (0.037)
Observations	1,545	1,708	1,503
R ²	0.012	0.004	0.017
Adjusted R ²	0.009	0.001	0.014
Residual Std. Error	0.498 (df = 1539)	1.154 (df = 1702)	0.472 (df = 1497)
F Statistic	3.818*** (df = 5; 1539)	1.333 (df = 5; 1702)	5.159*** (df = 5; 1497)

Notes: OLS models. *Wi-Fi autojoin* and *Updates via Wi-Fi* are indicator variables. Standard errors clustered at the individual level in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

C. Instructions and Screenshots

GlassWire App Screenshots and Description

The Android application used to monitor data consumption in the study is called *GlassWire* and is freely available on any Android app stores.

The application's main page reports the history of data consumption from both mobile and Wi-Fi networks, both in upload and download. The main page also reports the total data consumption (see screenshot below).



Screenshot of the main page of the application *Glasswire*. In this example, the total amount of data consumed is 68.5 Gb, 50.6 Gb of which comes from mobile (4/5G) network, and the remaining 17.9 Gb from Wi-Fi network.

The top menu of the application allows users to select the timeframe (e.g., week or month) and the source (Mobile and/or Wi-Fi, with the default screen being both).

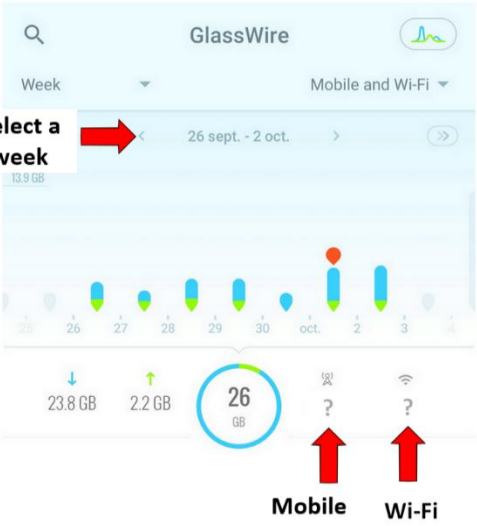
At the time of the experiment, the application did not allow users to set reminders about data consumption limits, nor to send push notifications. The latest updates have introduced some of these features.

Declaration Page

The screenshot below shows the declaration page coded in oTree (Chen et al. 2016) that participants in the experiment used to report their personal consumption of data in a given week. The screenshot below (translated from French) shows the declaration page of the first report (Report 0) in which participants were asked to report the consumption over the past three weeks. Before this page, participants were asked to download the application following our tutorial.

Now it's your turn!

Select the weeks indicated below on the application and report your Wi-Fi and Mobile data consumption.
Press "Next" as soon as you've finished. If you want to read the instructions again, press "Back".



Select a week

26 sept. - 2 oct.

23.8 GB

2.2 GB

26 GB

Mobile **Wi-Fi**

Week 3 oct. - 9 oct.

Mobile Data

Units

Wi-Fi Data

Units

Press "Next" when you're finished.

[Previous](#) [Next](#)

Pages shown exclusively in experimental conditions. a) information provided in the *Information* and *Goal setting* groups, b) goal setting and goal setting page where participants in the *Goal setting* group can set their weekly goals.

a)

Did you know that mobile networks produce up to 4 times more CO₂ than Wi-Fi networks?

According to your declaration of **Mobile** data consumption, you will produce:

2290g of CO₂/per month

If you had used **Wi-Fi** only with the same amount of consumed data, you will produce only:

536g of CO₂/per month

By **switching** to **Wi-Fi** network, your **CO₂ emissions will be reduced** by a quantity equivalent to:

32h electricity heating	
219 Km in High speed train	
2928h of LED bulb on	

As a reminder, here is your breakdown of last week's data consumption:
38% via Wi-Fi
62% via Mobile

Next

b)

My data consumption goal

You can set yourself a **Wi-Fi and Mobile data consumption goal for the coming week**.

Next Monday, you will be able to **compare your actual data consumption with your goals** to check whether you have achieved them.

As a reminder, here are your data consumption levels from the **last week** :

6,0 Gb via **Wi-Fi**

10,0 Gb via **Mobile**

Please, set here below your goals for the **coming week** :

Wi-Fi goal (Gb)	Mobile goal (Gb)
12	4
Your CO₂ emission calculator : You will produce 1052.3 g of CO₂ less than last week	

By clicking "Next", I make a commitment to myself to respect the consumption goals I have set.

Next

Feedback Page

Below is the page that participants in the *Goal setting* condition see after declaring past week's consumption (report) and before setting a new goal.

Your goals and consumption

Last week, you set your mobile and Wi-Fi data consumption goals. Based on last week's consumption report, you can now compare your goals with your actual consumption:

	Your consumption	Your goals	Difference
Mobile	8,0 Gb	9 Gb	-1,0 Gb
Wi-Fi	10,0 Gb	9 Gb	1,0 Gb

Results Mobile consumption : You have consumed 1.0 Gb **less** than your goal.

Results Wi-Fi consumption : You have consumed 1.0 Gb **more** than your goal.

Click "Next" to proceed.

Next

D. Initial and Final Questionnaires

In what follows, we report the questionnaires in the exact order that we administer throughout the whole study.

Initial Questionnaire (in Report 0)

- Knowledge about offloading: aims at assessing pre-experimental knowledge about offloading consequences to the environment. It includes also questions about the environmental consequences of transports. It was administered after data declaration, and before provision of information in the experimental conditions.
- General questionnaire: aims at understanding demographics, individuals' use of Wi-Fi and mobile as well as their habits.
- Big 5 questionnaire on personality traits
- Environmental concerns
- Time preferences

Final Questionnaire (in Report 5)

- General questionnaire (a subset of the questions of the general questionnaire of Report 0; in particular, only items about Wi-Fi use, device settings, mobile hotspot).
- Technology intention to use
- Reactance to interventions
- Knowledge about offloading
- Stated reactance to interventions

1. Knowledge about offloading and transports

Administered in the Initial Questionnaire and in the Follow-up Questionnaire; administered in Pure Control on the day of recruitment.

You need to compare 2 actions in terms of CO2 emissions produced.

- In your opinion, travelling 10 km by car (internal combustion engine) compared to travelling 10 km by TER (electric train) produces a quantity of CO2:
- In your opinion, travelling 10 km by plane compared to travelling 10 km by TGV (fast train) produces a quantity of CO2:
- In your opinion, consuming 100 GB of internet on a smartphone using the Wi-Fi network compared to consuming the same amount of GB on a mobile network (4G/5G) produces an amount of CO2:

Answers: Much lower, Lower, Equal, Higher, Much higher

2. General Questionnaire

Wi-Fi use

We now ask you to answer a short questionnaire.

- Do you have a Wi-Fi connection at home? Yes ; No
- How often do you use a Wi-Fi connection at home with your smartphone ? 1 (never) - 5 (always)
- On a scale of 1 to 5 (1 being “Very dissatisfied” and 5 “Very satisfied”) how satisfied are you with the Wi-Fi coverage in your usual home?
- On a scale of 1 to 5 (1 being “Very dissatisfied” and 5 “Very satisfied”) how satisfied are you with the speed of your Wi-Fi in your usual home?
- At home, is your Internet box switched on all the time? Yes ; No
- Does your internet box stay on when you are away for several days (vacations etc.)? Yes ; No
- On your Internet box, is Wi-Fi enabled? Never ; Only when I need it (or when someone else in my home needs it) ; All the time
- How often do you use a Wi-Fi connection outside your home (e.g. university Wi-Fi, work, public): 1 (never) - 5 (always)

Device Settings

- Is your smartphone set to automatically connect to Wi-Fi networks when it detects an accessible one? Yes ; No; I don't know
- Is your smartphone set to update exclusively via Wi-Fi? Yes ; No ; I don't know

Mobile Hotspot

- How often do you use your mobile as a “hotspot” (sharing mobile data) ? 1 (never) - 5 (always)

Mobile plan

- Do you have a 4G or 5G mobile plan? Yes / No
- If so, how many GB?

Preferences mobile over Wi-Fi

In general, when Wi-Fi is available (in a place other than your home), do you prefer to use your smartphone using Wi-Fi rather than the mobile network? Yes ; No

In general, you prefer to use Wi-Fi for:

- Download speed

- Signal coverage
- simplicity
- Security
- Not using data on my mobile plan

Others : [open text]

[Alternative if mobile chosen:]

In general, you prefer to use the mobile network for:

- Download speed
- Signal coverage
- simplicity
- Security

Others: [open text]

3. Big5 questionnaire

from Courtois et al. 2020 ([link](#)) (based on [Courtois et al. 2010](#))

You'll find a number of qualifiers that may or may not apply to you.

For each statement, indicate how strongly you agree or disagree with it:

1 - strongly agree, 2 - slightly disapprove, 3 - neither approve nor disapprove, 4 - approve a little, 5 - strongly approve

I see myself as someone who ...

1. Is reserved (6)
2. Tends to criticize others (2)
3. Works conscientiously (3)
4. Is relaxed, handles stress well (9)
5. Has a great imagination (20)
6. Sociable, outgoing (36)
7. Generally trusts others (22)
8. Tends to be lazy (23)
9. Easily anxious (39)
10. Has little interest in anything artistic (41)

Mean of: Extraversion : 6R, 36; Agreeability : 2R, 22; Conscientiousness : 3, 23R; Neuroticism : 9R, 39; Openness : 20, 41R.

4. Environmental concerns

Green self-identity from et al. Whitmarsh and O'Neill 2010

Please indicate the extent to which you agree with the following statements

Indicate the extent to which you agree or disagree with each statement:

1 - strongly disapprove, 2 - slightly disapprove, 3 - neither approve nor disapprove, 4 - approve a little, 5 - strongly approve

Questions:

1. I consider myself to be an environmentally conscious person.
2. I consider myself to be very concerned about environmental issues.
3. I would be happy to be considered as a person with an environmentally friendly lifestyle.
4. I would like my family or friends to see me as someone who cares about environmental issues.

5. Time discounting preferences

From Falk et al. 2023

Compared to others, are you a person generally willing to give up something today in order to benefit from it in the future, or are you not willing to do so?

Please use a scale from 0 to 10, where 0 means you're "absolutely not willing to give up anything today" and 10 means you're "very willing to give up something today".

You can also use the intermediate values to indicate where you stand on the scale.

6. Technology intention to use

Administered at the end of the experiment in all conditions

Intention to use the app in the future

Please indicate the extent to which you agree with the following statements.

For each statement, indicate how strongly you agree or disagree with it:

1 - no, with certainty, 2 - no, it's unlikely, 3 - don't know, 4 - yes, it's probable, 5 - yes, with certainty

1. I intend to continue using the GlassWire application in the future.
2. I intend to use the GlassWire application if:
 - technical information on CO2 emissions from Wi-Fi and mobile networks is added.
 - technical information on CO2 emissions from Wi-Fi and mobile networks and the possibility of setting a consumption target are added.

Frequency app checking

How many times a week have you checked the "GlassWire" application (beyond the time of reporting for the study)?

7. Stated Reactance to interventions

Administered at the end of the experiment only in Information and Goal setting conditions

For each statement, enter the number indicating how strongly you agree or disagree with it:

1 - strongly disapproves, 2 - slightly disapproves, 3 - neither approves nor disapproves, 4 - approves a little, 5 - strongly approves

General reactance

[if Information] : I found the technical information on CO2 emissions from Wi-Fi and mobile networks produced by my uses interesting.

[if Goal setting] : I found it interesting to set myself a target for the CO2 emissions from Wi-Fi and mobile networks produced by my usage.

Threat to freedom

[if Information] : I felt manipulated by the technical information on CO2 emissions from Wi-Fi and mobile networks produced by my usage.

[if Goal setting] : I felt manipulated by setting myself a target for the CO2 emissions from Wi-Fi and mobile networks produced by my usage.

E. Follow-up Questionnaire

Administered 6 months after the end of the main experiment.

- Knowledge about offloading (see *Initial and Final questionnaire*)
- Whether subjects kept the app, and, if yes, their reported frequency of use
- Green identity (see *Initial and Final questionnaire*)
- Empirical and Normative expectations, and Personal norms

The present work only focuses on the results of the knowledge about offloading. The other questions are part of a future separate study. We reported below the full questionnaires.

Keep the app and frequency

Have you kept the "GlassWire" application since the end of the "Android" study?

Answers: yes, no, I don't know, I don't want to answer

{if yes} How many times a week do you check the GlassWire application?

Empirical Expectations

You can earn an extra €12.50 if you answer this question correctly.

Guess the percentage of people who answered "Yes" to the previous question (previous question: "In recent weeks, have you used mobile data to watch streaming video (movies, social networks, video calls, etc.) when Wi-Fi was available?").

Your earnings: You'll be entitled to an additional win of €12.50 if you guess the true percentage or if your estimate is five units lower or higher than that percentage. This means that, if you are drawn at the end of the study, you can earn €25 for completing the study + €12.50 if you guess the percentage in this section.

Normative expectations

You can earn an extra €12.50 if you answer this question correctly.

Consider the following scenario:

Individual A decided to watch streaming videos (movies, social networks, video calls, etc.) on his smartphone using the mobile network (4/5G) while a Wi-Fi connection was available.

We ask you to decide whether this action could be considered "socially appropriate" (and "consistent with moral or proper social behavior") or "socially inappropriate" (and "inconsistent with moral or proper social behavior").

By "socially appropriate", we mean behavior that most people consider "correct" or "ethical". Another way of understanding what we mean is that if an individual makes a socially inappropriate choice, someone else might disagree with them about it. This question doesn't ask for your personal opinion, but your assessment of society's opinion. We ask you to answer as accurately as possible.

Your earnings: At the end of the study, we will determine the most frequent answer given by the other participants. If you are drawn at random, you will win an additional €12.50 if your answer is identical to the most frequent answer given by the other participants in the study.

If, on the other hand, you don't give the same answer as most of the other participants, you won't be entitled to the extra €12.50 in this task. In other words, to qualify for the extra earnings in this task, you must provide the answer you think is most frequent among all study participants.

Personal Norms

Unlike the previous part, in this part of the study you will assess whether the same action described above is appropriate or not, according to your own opinion and independently of the opinion of others.

Appropriate” behavior refers to behavior that you personally consider to be ‘correct’ or ‘moral’.

This question asks for your personal opinion, independent of the opinions of others.

We ask you to answer as accurately as possible, and to give your honest opinion. There are no right or wrong answers; you will receive no additional payment for your answers in this section.

F. Power analysis and sample justification

To make sure that our sample size allows us to detect small changes in our variables of interest, we ran a power analysis to estimate the required sample to estimate an effect of interest. To do so, we ran several Monte-Carlo simulations using the R package *rsim*, to estimate the main model in Eq. 1. We assumed a minimum detectable of 0.1 standard deviations (i.e., a Cohen's $d=0.10$, a small effect, equivalent to a raw change in % *Wi-Fi* of 2.5% based on pilot data) and sequentially varied the number of observations per experimental condition. To obtain a power of 80%, the required sample is of 240. All materials are available at the following link:

https://researchbox.org/1325&PEER_REVIEW_passcode=TZXHQ

G. Pre-registrations and replication material

This study contains two pre-registrations. The first pre-registration (https://aspredicted.org/4J6_MHC) concerns the main experiment (Wave 1) and contains the following hypotheses tested in the results section of the main text. The word soft commitment is used interchangeably with goal setting:

HP1: self-monitoring via the app reduces on average mobile data consumption;
HP2: provision of technical information reduces mobile data consumption;
HP3: provision of technical information combined with soft commitments (goal setting) reduces mobile data consumption.

To test *HP1*, we compared pre-experimental level of consumption vs. experimental levels (after the download of the app). The results are discussed in Section 3.3 and the regression estimates are reported in Table A11.

To test *HP2* we compare % *Wi-Fi* in *Baseline* vs. *Information* condition. The results are discussed in Section 3.1 and regression estimates are reported in Table 2 and Table A4-A5.

To test *HP3* we compare % *Wi-Fi* in *Information* vs. *Commitment (Goal setting)* condition. The results are discussed in Section 3.1 and regression estimates are reported in Table A7.

The second pre-registration (https://aspredicted.org/1ZB_FHF) concerns the “*Follow up*” questionnaire (see Figure 1 in the main text) and includes the following tested hypotheses:

HP4: we expect that subjects in our *Information* and *Commitment (Goal setting)* conditions recall the information provided in the experiment about the advantages and consequences of offloading and mobile vs. *Wi-Fi* pollution.

To test *HP4*, we administered the items used to assess knowledge about offloading, 6 months following the main study, and tested any difference across waves using a difference-in-difference approach using the *Baseline* condition as benchmark. Results are discussed in Section 4 and reported in Table A12.

Further details and content not included in the pre-registration

To preserve transparency of our work, in this section, we discuss and motivate some of the contents included in the manuscript but not included in our pre-registrations, as well as any decisions that could potentially be considered a deviation from initial plans.

Outliers. Although we did not pre-register any a-priori rule to exclude observations, in the manuscript, we excluded 21 observations (0.2% of total) as the related consumption measures are impossible to observe in reality (for example, this can be due to misreporting, mistake in selecting Mb or Gb, or purposefully wrong answer given by the participants; see Section 2.1). These observations were detected by eyerolling the dataset, and by calculating z-scores of mobile and Wi-Fi consumption and classifying them as outliers (threshold set to 2). These observations have been excluded from all our analyses.

Pure Control inclusion. To provide a better empirical test of our HP1, we recruited a separate sample of participants right after the end of the main study, that we called *Pure Control* condition in the main text. This data collection was not initially included in the first pre-registration as we were not initially sure to be able to conduct an extra condition and ensure participants in our subject pool. However, these results aim to complement the pre-registered analyses reported in the manuscript (Section 3.3).

Additional hypotheses in Wave 2. The second pre-registration includes some additional hypotheses that will be the object of a follow-up manuscript and have therefore been excluded from the present one. These are:

HP5: we expect that social and personal expectations about preferring Wi-Fi to mobile are stronger in our nudge experimental conditions.

HP6: we expect green identity to become higher following nudges

HP7: we expect that our nudges interventions (Information and Commitment, see previous study, Wave 1) lead to higher rates of application adoption (technology adoption).

Reporting additional regression models. Our pre-registered analyses initially relied on linear regression models for hypothesis testing. However, because our key variable of interest (% Wi-Fi) is censored, linear regression alone may lead to assumption violations. To address this, we incorporated Tobit models, which are specifically designed for censored data. As a result, we present Tobit model results for % Wi-Fi in the main text (e.g., Table 2) while providing supplementary linear regression analyses in the Appendix (e.g., Table A7) for comparison.

H. Co-Design sessions

Using a separate sample of students (N=60), we organized a session to co-design the information page used in the experiment.

At the beginning of these sessions, we elicited their offloading knowledge using the question we implemented in the actual experiment. We also asked their empirical expectations (i.e., their best guess) of the percentage of Wi-Fi among their peers in the same session.

The second sets of questions asked to guess the most impactful message that according to the respondent would have produced a change in behavior among their peers, using two possible scenarios (low consumption of data – X=25Gb, and high consumption of data – X=100Gb). The value Y changes according to the scenario.

Q1. On a scale of 0 to 5 (where 0 means “no impact”; and 5 “very strong impact”), rate the ability of the following messages to shift mobile data consumption to Wi-Fi.

- The mobile network consumes 4 times more energy than Wi-Fi
- {X} GB consumed using Wi-Fi rather than Mobile generates less CO2 equivalent to {Y} km by plane
- {X} GB consumed by using Wi-Fi rather than Mobile generate less CO2 equivalent to {Y} Km by TGV (high-speed train)
- {X} GB consumed using Wi-Fi rather than Mobile generate less CO2 equivalent to {Y} Km by bus
- {X} GB consumed using Wi-Fi rather than cell phones produces less CO2 equivalent to {Y} hours of heating (for an 80m² apartment)
- {X} GB consumed using Wi-Fi rather than Mobile generates less CO2 equivalent to {Y} hours of LED lighting

Q2. Of the messages you evaluated, please indicate the 2 most impactful and most likely to switch mobile data consumption to Wi-Fi.

{X} GB consumed using Wi-Fi rather than Mobile produce less CO2 equivalent to :

- {Y} km by plane
- {Y} km by TGV
- {Y} km by bus
- {Y} hours of heating (for an 80m² apartment)
- {Y} hours of LED bulb lighting

In a second set of questions, we asked to express their best guess on which of the described interventions would likely bring about a change in offloading levels among peers.

Q3. We're now going to present you with a series of possible solutions to encourage you to switch your mobile data consumption to Wi-Fi.

On a scale of 0 to 5 (where 0 means “no impact” and 5 “very high impact”), judge the impact of each solution.

- Compare your own mobile (4/5G) and Wi-Fi data consumption with that of others.
- Commit to reducing your mobile data consumption in favor of Wi-Fi on a public website.
- Make a private goal (via an application, for example) to reduce your mobile data consumption in favor of Wi-Fi.